

Alpha Momentum and Alpha Reversal in Country and Industry Equity

Indexes

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Author's Note

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Abstract

Do past alphas predict future country and industry returns? Examination of equity indexes from 51 stock markets between 1973 and 2018 allows us to demonstrate new return patterns in the cross-section of country and industry returns. Past short-term (long-term) alphas positively (negatively) predict future returns. These phenomena can be translated into effective international equity allocation strategies, producing economically and statistically significant raw and risk-adjusted returns. The profitability is robust to many considerations, including alternative alpha models, the effect of trading costs, different holding periods, or subsample and subperiod analyses. Also, the alpha momentum and reversal subsume their return-based counterparts. Finally, the alpha-based patterns are particularly pronounced following bull markets and across the markets characterized by high arbitrage constraints, supporting the behavioral explanation of the anomalies.

Keywords: alpha momentum, alpha reversal, international investment, country momentum, country reversal, industry momentum, industry reversal, asset pricing, the cross-section of returns return predictability, equity anomalies,.

JEL codes: G12, G15.

1. Introduction

A past alpha is a powerful predictor of future stock returns. Hühn and Scholz (2018) show that a short-term alpha is positively correlated with future payoffs: the higher the alpha, the better the future performance. On the other hand, Horenstein (2018) offers a “betting against alpha” strategy, implying that the stocks with high long-term alpha underperform those with low long-term alpha. The major aim of this study is to examine whether the recently documented alpha-based return patterns for stocks also exist for countries and industries. We show that the long-term and short-term alphas offer a very effective return predictive variable which could be used for international asset allocation strategies across countries and industries.

The momentum effect, which is the tendency of stocks with low (high) past returns to underperform (overperform), is one of the most pervasive and omnipresent anomalies ever discovered. Traced back to Jegadeesh and Titman (1993), it has been documented in international equities (Chui, Titman, and Wei 2010), country and sector indexes (e.g., Moskowitz and Grinblatt 1999; Bhojraj and Swaminathan 2006), corporate and government commodities, bonds, and forex exchange (Asness, Moskowitz, and Pedersen 2013). More importantly, momentum influences not only the total returns on stocks, but also return components after disaggregation with multifactor models. Ehsani (2017) shows that the momentum phenomenon affects the performance of asset pricing factors driving equity returns. Gutierrez and Pirinsky (2007), Blitz, Huij, and Martens (2011), and Blitz, Hanauer, and Vidojevic (2017) deliver convincing evidence that a highly effective momentum strategy could also be based on return residuals derived from asset pricing models. Another stream of research documents a reversal effect in asset returns. De Bondt and Thaler (1985), and Daniel, Hirshleifer, and Subrahmanyam (1998) indicate that investors who are overconfident in their information overreact to past information and move the prices away from

fundamentals. However, in the long-term, investors learn that their initial judgments about prices were biased and the mispricing is corrected causing prices to revert to fair values—the reversal effect. Hühn and Scholz (2018) and Horenstein (2018) offer momentum and reversal strategies that are based on alpha rather than raw returns and show that alpha-based strategies yield considerable returns.

The intuition behind using alpha-based strategies can be explained as the following. Standard momentum and reversal strategies are subject to systematic risk and an important portion of the returns attributed to these strategies can be driven by factor risks. Employing a trading strategy that is based momentum and reversal signals based on alphas, adjusts the past returns for risk. In consequence, it disentangles the momentum and reversal effects from the effects of risk factors. Moreover, an alpha-based strategy diverges from its standard counterpart in the sense that it benefits from mispricing in assets by taking a long position in undervalued assets with positive alphas and a short position in overvalued ones with negative alphas.

Hühn and Scholz (2018) argue that the alphas from asset pricing models are positively correlated with future returns.¹ Focusing on U.S. equities, they estimate a three-factor model alpha based on the trailing 12-month period and document that higher alphas correspond to better future payoffs. Using a markedly longer window of five years to estimate alphas, Horenstein (2018) finds that realized alphas are negatively related to future stock performance.

How can this mixed evidence be resolved? Considering the difference in the alpha estimation period, we assume that the discrepancy could be reconciled within a behavioral framework. Taking the classical momentum effect as a parallel example, there is no contradiction

¹ Arithmetically, the concept of alpha momentum is closely related to the residual momentum, but the derivation of the return predictive signal requires shorter time-series of data.

between the short-term continuation in returns and long-run reversal. Although momentum strategies typically work with short-term sorting periods, from 1 to 12 months, the contrarian momentum strategies use sorting periods of 3 to 5 years. Moreover, these patterns are consistent with the overreaction hypothesis put forward by DeBondt and Thaler (1985, 1987), formalized later by DeLong, Shleifer, Summers, and Waldmann (1990), and explored in many further studies, including Daniel, Hirshleifer, and Subrahmanyam (1998), Barberis, Shleifer, and Vishny (1998), Hong and Stein (1999), Balvers and Wu (2006), and Wu (2011). A series of momentum profits (losses) may lead to a positive (negative) long-run return and a subsequent overvaluation (undervaluation), which is corrected toward the intrinsic value through a reversal mechanism in the long term. Consistent with that, Balvers and Wu (2006), Jegadeesh and Titman (2001), and Lee and Swaminathan (2000) found that certain momentum portfolios exhibit eventual mean reversion. Thus, momentum and reversal effects, considered jointly, form a sort of “behavioral cycle” observed in equity markets.

We hypothesize that an analogous behavioral explanation may hold for the conflicting evidence regarding past alphas presented by Hühn and Scholz (2018) and Horenstein (2018). In consequence, we expect short-term alphas to be positively correlated with future returns and long-run alphas to be negatively correlated. In other words, we expect the assets to exhibit short-run alpha momentum and long-term alpha reversal.

To test this conjecture, we investigate a sample of 51 country indexes and 887 industry indexes for the years 1973–2017. We follow the framework of earlier studies that research momentum and reversal patterns at country (Bhojraj and Swaminathan 2006; Balvers and Wu 2006; Bornholt and Malin 2010) and industry levels (Moskowitz and Grinblatt 1999; Shen, Szakmary, and Sharma 2005; Du and Denning 2005; Bornholt, Gharaibeh, and Malin 2015), but

integrate and extend their scope to obtain the most comprehensive sample ever investigated. We calculate two distinct return predictive signals: the *alpha momentum*, which is the alpha derived from a 12-month trailing estimation period, and the *alpha reversal*, which is the alpha derived from the 60-month period with the 12 most recent months skipped in order to avoid the overlap with the momentum signal. Subsequently, we apply an array of cross-sectional and time-series tests to investigate the role of past alphas in predicting future returns. We conduct Fama-MacBeth (1973) regressions as well as examine the returns of portfolios formed on past alphas from one-way and two-way sorts. We are also interested in the relationship between alpha-based and price-based momentum and reversal signals, to see which is the underlying driver of the other. Finally, the outcomes are verified with a battery of robustness tests, including controlling for other return predictive variables, accounting for trading costs, and using alternative weighting methods, subsample and subperiod analyses, and factor spanning tests.

The principal findings of the study can be summarized as follows. First, we document a strong and robust short-term alpha momentum effect and long-run reversal effect in industry and country indexes. The 6–12 month trailing alpha positively predicts future returns. On the other hand, the 6–120 month trailing alpha lagged by 12 months is consistently negatively correlated with future returns. The results hold in both cross-sectional and portfolio-based time-series tests. The alpha momentum in particular is a powerful return pattern, being robust to many considerations, such as the impact of trading costs and alternative holding periods, different portfolio weighting methods, or application in various subperiods and subsamples, including developed and emerging equities. The equal-weighted strategy going long in the quintile countries (industries) with the highest 12-month alpha, simultaneously shorting the quintile with the lowest

alpha, delivers a three-factor model alpha of 0.92% (1.41%) per month with a Sharpe ratio of 0.58 (0.93), nearly doubling (tripling) that of the market portfolio.

Second, we document that alpha momentum and reversal are not just new manifestations of simple price momentum and reversal anomalies—these are independent asset pricing phenomena that actually constitute underlying drivers of the price momentum and reversal. To investigate the relationship between the return-based and alpha-based anomalies, we conduct three separate examinations: multiple cross-sectional regressions, times-series spanning tests, and evaluations of portfolios from two-way dependent sorts. The alpha momentum effect subsumes the price momentum entirely. In nearly all variants and specifications of our tests, the alpha momentum fully explains the price momentum, but the price momentum does not explain the alpha momentum. For the long-run reversal effect, the relationship is less striking, but still consistent. Once we control for the role of return reversal, the alpha reversal remains significant only in selected market segments or some of the tests. However, the return reversal never retains its significance when we account for the role of alpha reversal. In other words, the alpha reversal deals well with explaining the return reversal effect.

Third, we show that our results support the behavioral explanation of the alpha momentum and reversal in international equity markets. The behavioral story of equity anomalies indicates that they result from investor irrationality that cannot be easily arbitrated away. Consistent with this, we find that both the alpha momentum and reversal phenomena are particularly pronounced within the market segments characterized by elevated limits to arbitrage, namely, small markets and industries with high idiosyncratic volatility and low correlation with global equity market portfolios. Also, in line with the arguments of Cooper et al. (2004), the alpha momentum and

reversal are remarkably significant following bull markets, accentuating the overconfidence roots of the alpha momentum and reversal effects.

The study aims at contributing in two major ways. First, it reconciles the earlier evidence of the alpha momentum and reversal phenomena documented by Hühn and Scholz (2018) and Horenstein (2018), respectively. We show that these are two independent effects derived from different estimation periods, and are consistent with the overreaction hypothesis that explains short-term momentum and long-term reversal.

Second, this is the first study to demonstrate the alpha momentum and reversal effects in the universe of country and industry indexes. Earlier studies of the predictive abilities of alphas (Hühn and Scholz 2018; Horenstein 2018) or even of the partially related residual momentum (Gutierrez and Pirinsky 2007; Blitz, Huij, and Martens 2011; Chaves 2012; Blitz, Hanauer, and Vidojevic 2017) were conducted at the stock level. Meanwhile, the country- and index-level studies focused solely on the regular momentum and reversals (e.g., Moskowitz and Grinblatt 1999; Du and Denning 2005; Shen, Szakmary, and Sharma 2005; Balvers and Wu 2006; Bhojraj and Swaminathan 2006; Bornholt and Malin 2010, 2013; Bornholt, Gharaibeh, and Malin 2015) or alternatively on its technical modifications, such as the volatility-adjusted momentum (e.g., Dudler, Gmur, and Mahmud 2015). Moreover, we further examine the relationship between alpha-based and return-based strategies and show that alpha-based strategies subsume the return-based strategies. Thus, we shed light on the potential sources of the classical return momentum and reversal effects.

From a practical perspective, by offering the alpha momentum and reversal strategies, we also expand the currently known list of return predictive signals for international equity indexes, including the size (Keppler and Traub 1993; Keppler and Encinosa 2011), valuation ratios

(Macedo 1995; Kim 2012), and systematic and idiosyncratic risks (Frazzini and Pedersen 2014; Bali and Cakici 2010; Umutlu 2015; Zaremba 2016), in addition to the momentum and reversal effects.

The rest of the article is structured as follows. Section 2 outlines the motivation and theoretical basis for our study. Section 3 presents the data and variables used in the study. Section 4 discusses the calculation of the alpha-based variables in this study. Section 5 shows and discusses the basic cross-sectional relations. Section 6 presents and analyzes the portfolios formed on past alphas. Eventually, Section 7 concludes the study.

2. Motivation and Theoretical Basis

A stream of research documents that a considerable portion of the profits from standard momentum and reversal strategies may be due to exposure to systematic risk. For instance, Grundy and Martin (2001) and Chordia and Shivakumar (2002) show that momentum portfolios have high betas; Fama and French (1993, 1996) provide a risk-based explanation for long-term reversal in the context of the value-growth paradigm; and Wu and Li (2011) find that value and growth properties of stock explain long-term reversal. However, it is also shown that risk factors cannot entirely predict the momentum and reversal returns (Moskowitz and Grinblatt 1999; Griffi, Ji, and Martin 2003; Cooper, Gutierrez, and Hameed 2004; De Bondt and Thaler 1987; Dissanaike 1997). In this study, we focus on momentum and reversal strategies that are based on risk-adjusted returns, i.e., alphas, which are isolated from systematic effects. Following an alpha-based strategy, one invests in undervalued stocks with positive alphas and short sells overvalued stocks with negative alphas. This is the well-known active portfolio management strategy that aims to benefit from mispricing. Put differently, alpha-based strategies exploit mispricing in assets. In this regard, alpha

momentum and alpha reversal strategies are distinguished from their standard counterparts, which are subject to factor risks and produce signals that are not directly linked to mispricing.

If mispricing is not corrected in the short run, one observes the alpha momentum effect—positive (negative) alphas continue to be positive (negative). Once the mispricing is corrected in the long run, alphas revert to zero and the prices reflect the fair values. One explanation for the delayed correction of alphas is the overreaction of investors to recently released news (De Bondt and Thaler 1985; Daniel, Hirshleifer, and Subrahmanyam 1998; Hong and Stein 1999). According to the overreaction hypothesis, investors are overconfident in their information and they overreact to past information. This pushes prices to deviate from fundamentals, resulting in mispricing and thus non-zero alphas. Momentum investors who buy after price increases and sell after price declines cause an overshooting effect and exacerbate mispricing by further moving the alphas away from zero. So, non-zero alphas get larger in absolute terms in the short run and create alpha momentum. When arbitrageurs are strong enough to correct mispricing, non-zero alphas revert to zero in the long run, which we refer to as alpha reversal. In short, overreaction causes short-run momentum and long-run reversal.

An implication of the overreaction hypothesis is that momentum should be greater in up markets. Daniel, Hirshleifer, and Subrahmanyam (1998) argue that when investors' overconfidence in their private information is combined with self-attribution bias, they react asymmetrically to public news depending on whether that public news confirms or denies their private information. Upon the arrival of public news supporting their private information, investors attribute more value to their skills than they deserve. When public news contradicts their private information, investors attribute this conflict to external factors rather than accepting the failure. As a result, investors become more overconfident after confirming public news and overreact,

stimulating higher momentum in the short run. Cooper, Gutierrez, and Hameed (2014) focus on the implications of the overreaction hypothesis of Daniel, Hirshleifer, and Subrahmanyam (1998) in different market states. They argue that overconfidence is accentuated in bull markets because aggregate investors in long positions perceive an overall increase in market prices as the public news confirming their private information and as affirmation of their investment skills. If overconfidence increases after markets rise, then bull markets should be characterized by higher levels of overreaction producing stronger short-run momentum.

Another implication of the overreaction hypothesis is that momentum effects should be more pronounced for assets with high idiosyncratic volatility. Idiosyncratic volatility can be interpreted as a proxy for firm-specific information. So, if momentum is caused by the overreaction of investors to firm-specific information, as suggested by De Bond and Thaler (1985, 1987) and Daniel, Hirshleifer, and Subrahmanyam (1998), then momentum effects should be stronger for assets with high idiosyncratic volatility. Supporting this view, Heidari (2015) theoretically shows that idiosyncratic volatility is positively associated with overreaction. In accordance with the implications of this theory, he also provides empirical evidence showing that the source of momentum profits is investor overreaction.

At a later stage of this article we demonstrate that our findings of alpha momentum in the short term and alpha reversal in the long term are consistent with the investor overreaction hypothesis. Moreover, our findings of more pronounced momentum effects in bull markets and among assets with high idiosyncratic volatility are also in line with the implications of the overreaction hypothesis. Overall, our results indicate that alpha momentum and alpha reversal effects have behavioral origins.

3. Data and Variables

This study is based on accounting and return data on international equity indexes sourced from the Thomson Reuters Datastream. The research relies on monthly observations from a sample period that runs from January 1973 to May 2018. We conduct our analyses in two different samples: country indexes and industry indexes. The individual country indexes cover 51 markets, including both developed and emerging ones. They are listed along with their basic properties in Table A1 of the Online Appendix. On the other hand, the industry indexes track industry portfolios in particular countries. These are formed according to the Industry Classification Benchmark (ICB). The equity market is split into 19 different industries: automobiles and parts, basic resources, banks, construction and materials, chemicals, food and beverage, financial services, industrial goods and services, healthcare, insurance, oil and gas, media, real estate, personal and household goods, technology, utilities, travel and leisure, telecommunications, and retail.² Naturally, not all the industries are present in every country, so in total we obtain 887 industry indexes. An overview of the sample of industry indexes is presented in Table A2 of the online Appendix.

All the data are initially collected in local currencies and then converted to U.S. dollars to form a pooled international sample. Consistent with this approach, we utilize the three-month T-bill rate obtained from French's (2018) representation of the risk-free return.

Finally, to assure the quality of the data, we delete the index-month observations with returns below -98% or exceeding 500%. Also, to overcome practical problems with investing in small industries in developing countries, we exclude the index portfolios with beginning-of-the-

² Further information on the ICB classification is available at www.icbenchmark.com.

month market capitalization below USD 100 million. In total, our sample of country (industry) returns includes 19,692 (292,645) monthly observations.

For the cross-sectional and time-series tests in the study, we calculate a few basic variables. The market value, MV_t , is the natural logarithm of the total stock market capitalization of the companies in the index portfolio in month $t-1$. The earnings-to-price ratio, EP_t , is the trailing 12-month profit of the companies in the index, weighted according to the index rules and divided by the total stock market capitalization of the index, both at time $t-1$. Importantly, we use accounting data on an “as available” basis to avoid look-ahead bias. $BETA_t$ and $IVOL_t$ are the stock market beta and idiosyncratic volatility from the CAPM model estimated from the monthly returns in the period $t-60$ to $t-1$. $SKEW_t$ is the product-moment skewness of the return distribution estimated through the months $t-24$ to $t-1$. $SEAS_t$ represents cross-sectional seasonality and is calculated as the average return in the same calendar month through the months $t-240$ to $t-1$, as available. The momentum variable, MOM_t , is computed as the total return on an index in months $t-12$ to $t-1$. Finally, the reversal variable, REV_t , is the total return on an index in months $t-60$ to $t-13$.

4. The Alpha Momentum and Alpha Reversal Signals

This paper focuses on the relationship between past alphas and future returns. We start our examinations with the simple CAPM model (Sharpe 1964), but for robustness we also examine three multifactor models following Fama and French (1993), Carhart (1997), and Asness, Moskowitz, and Pedersen (2013). The four models are represented by Equations (1), (2), (3), and (4), respectively:

$$R_{i,t} = \alpha_i + \beta_{MKT,i}MKT_t + \varepsilon_{i,t} \quad (1)$$

$$R_{i,t} = \alpha_i + \beta_{MKT,i}MKT_t + \beta_{SMB,i}SMB_t + \beta_{HML,i}HML_t + \varepsilon_{i,t} \quad (2)$$

$$R_{i,t} = \alpha_i + \beta_{MKT,i}MKT_t + \beta_{SMB,i}SMB_t + \beta_{HML,i}HML_t + \beta_{MOM,i}MOM_t + \varepsilon_{i,t} \quad (3)$$

$$R_{i,t} = \alpha_i + \beta_{MKT,i}MKT_t + \beta_{HML,i}HML_t + \beta_{MOM,i}MOM_t + \varepsilon_{i,t} \quad (4)$$

where $R_{i,t}$ is the monthly excess return on the analyzed portfolio i ; $\beta_{MKT,i}$, $\beta_{SMB,i}$, $\beta_{HML,i}$, and $\beta_{MOM,i}$ are measures of exposure to MKT_t (market excess return), SMB_t (small minus big), HML_t (high minus low), and MOM_t (momentum, or up minus down) risk factors, respectively; $\varepsilon_{i,t}$ is the error term; and the α_i intercept represents the *alpha* or the average abnormal return.³ In particular, Equation (2), which represents the basic model, is commonly employed in stock-level studies on alpha momentum (Horenstein 2018; Hühn and Scholz 2018) or other types of momentum requiring residualization of returns (Gutierrez and Pirinsky 2007; Blitz, Huij, and Martens 2011; Blitz, Hanauer, and Vidojevic 2018; Zaremba, Umutlu, and Maydybura 2018).

To obtain the alpha score, we loosely follow Blitz, Huij, and Martens (2011) and Blitz, Hanauer, and Vidojevic (2018) and scale the alphas α_i from Equations (1)-(4) by the volatility of returns:

$$\alpha_{S,i} = \frac{\alpha_i}{\sqrt{\sum_{j=1}^n (R_{i,t-j} - \bar{R}_i)^2 / n}}. \quad (5)$$

As in Blitz, Hanauer, and Vidojevic (2018), this adjustment leads to a slight improvement in the risk-adjusted performance, but the results of this paper do not hinge on it. At a later stage of the paper we demonstrate that our results are qualitatively robust to the lack of this adjustment and, in fact, the non-adjusted alpha signal frequently generates raw and risk-adjusted returns on long-short alpha-based portfolios of a marginally larger magnitude.

³ Importantly, our approach to obtaining alpha departs from Hühn and Scholz (2018), who rely on daily data. Monthly returns better cope with microstructural issues and different session timings around the world; they also allow better computation efficiency when dealing with comprehensive datasets.

The detailed procedures for the formation of the asset pricing factors used in these models are reported in Tables A3 and A4 of the online Appendix, along with basic return characteristics. Also, Figure A1 in the Appendix presents the cumulative returns on factor portfolios. Importantly, to assure consistency, when examining the country and industry universe we form separate factors based on country and industry data, respectively. As illustrated in Table A4 and Figure A1, the performance of country-based and index-based factors differs only marginally.

In general, when calculating the factors, we apply the standard procedures originating from Fama and French (1993). Nevertheless, we introduce three minor changes determined by the index-level character of our study. First, we follow Zaremba (2015) and calculate the asset-pricing factors based on index-level data to ensure consistency with the rest of our study. Second, in the computation of the *HML* factor, in line with Umutlu (2015), we substitute *EP* for the book-to-market ratio, which is motivated by better data availability. Finally, contrary to the usual procedures in the examinations of individual companies (for instance, Fama and French [2012]), when forming the standard momentum factor (*MOM*), we do not drop the most recent month's payoff. This is because of lack of any evidence of the short-run reversal at the level of countries, and the indirect evidence even suggests short-run continuation rather than the reversion (Blitz and van Vliet, 2008; Simpson, Giudici, and Emery, 2011).⁴

⁴ Table A5 in the Online Appendix exhibits a few simple cross-sectional regressions shedding light on the predictive abilities of the variables underlying the factor portfolios. Table A6 uncovers the performance of asset pricing models. Following the suggestions of Balvers and Stivers (2018), we consider the ability to explain both the cross-sectional and time-series variations in returns. Regarding the time-series variation, the market factor seems to be of crucial importance, and adding the further factors plays only a minor role. However, when it comes to explanatory power over the cross-sectional returns, considering the additional factors plays a crucial role, markedly improving the models' performance.

Previous financial studies relied on various window lengths to estimate the historical alpha. In particular, Hühn and Scholz (2018) used a 12-month estimation period, whereas Horenstein (2018) employed a 5-year window. To investigate the relationship between the length of the look-back period and the future returns, we examine a broad range of window lengths ranging from the trailing 6 months to 120 months. Furthermore, to control for the influence of the most recent performance, we also research these periods lagged by 12 months.

In addition to the wide-scale overview of various estimation periods, we choose two specific alpha-based variables as the basis for more in-depth examinations, namely: the *alpha momentum* (*AMOM*) and the *alpha reversal* (*AREV*). The estimation periods for these variables have been chosen arbitrarily based on similar studies of simple price momentum and reversal effects in equity markets. Importantly, the signals estimated during these periods also display a strong correlation with future returns in our study. The *AMOM* in month t is measured as the volatility-adjusted alpha estimated during months $t-12$ to $t-1$. The 12-month window avoids the influence of monthly calendar effects in returns and is also consistent with the earlier studies of, e.g., Hühn and Scholz (2018). Importantly, contrary to Hühn and Scholz (2018), we do not abandon the most recent month due to lack of evidence of the short-run reversal in index-level returns (Blitz and van Vliet, 2008; Simpson, Giudici, and Emery, 2011).

The *AREV* variable in month t is the volatility-adjusted alpha estimated during a 60-month period with the 12 most recent months skipped, i.e., through the months $t-60$ to $t-13$. Following, e.g., Jacobs (2015), the most recent 12 months are discarded in order to avoid the overlap with the momentum measure.

5. Basic Cross-Sectional Regressions

We begin our investigations by checking the cross-sectional relations following Fama and MacBeth (1973):

$$R_{i,t} = \beta_{0,t} + \sum_{j=1}^J \beta_{j,t} K_{i,t-1} + \varepsilon_{i,t}, \quad (6)$$

where $R_{i,t}$ denotes the excess return on the index portfolio i in month t , $\beta_{0,t}$ and $\beta_{j,t}$ represent regression coefficients, and $K_{i,t-1}$ indicate various return predictive variables. In the first pass, we apply the simple Fama-MacBeth regressions to the excess returns on indexes, using only one independent variable: the CAPM alpha. We aim at a quick scan of the abilities of the alphas derived from different estimation periods to predict returns in the cross-section.

Figure 1 reports the average regression coefficients along with the corresponding t -statistics. An eyeball test reveals an interesting pattern in the dependence of the returns on the past alphas. Panel A of Figure 1 indicates that the regression coefficients are positive in the short term, reaching their peak of 0.0129 (0.0130) for countries (industries) around the 12th month. The significance levels (Panel C of Figure 1) are high and the corresponding t -statistics for the 12th month amount to 4.26 (5.32) for countries (industries). This observation matches with the stock-level findings of Hühn and Scholz (2018) that the short-term alpha positively predicts future returns. Conversely, when the estimation period is extended, the cross-sectional correlations turn visibly negative during around the 60th month for both countries and industries. In other words, the long-run alphas negatively predict the future returns. This observation, in turn, corroborates the conclusions of Horenstein (2018) that it makes sense to “bet against alpha.”

[Figure 1 about here]

Panels B and D of Figure 1 provide further insights into the issue of negative correlation. We apply the Fama-MacBeth regressions after discarding the most recent 12 months from the alpha

estimation period. The results clearly point to the conclusion that any positive correlation with alpha is driven solely by the most recent calendar year. Any alpha based on an earlier estimation period is negatively correlated with future returns, and the economic significance of this phenomenon is substantial. The regression coefficients are particularly high for the five- to eight-year formation periods, with the slope estimates frequently exceeding -0.02 (-0.03) for countries (industries) and the corresponding t -statistics surpassing -3.0 (-4.0). Summing up, whereas the short-term alpha is positively related to returns, the long-term alpha works in the opposite way: the higher the alpha, the lower the future return. This phenomenon resembles the momentum and reversal cycle in country equity indexes documented by Richards (1997), Connolly and Stivers (2003), and Balvers and Wu (2006), among others.

Table 1 presents the results of further investigations using Fama-MacBeth regressions. In particular, we are interested in whether the alpha effects hold after controlling for popular return predictors, including beta (*BETA*), market value (*MV*), earnings-to-price ratio (*BM*), idiosyncratic volatility (*IVOL*), skewness (*SKEW*), and cross-sectional seasonality (*SEAS*). Henceforth, we limit our attention to only two alpha-related variables: *AMOM* and *AREV*. In our default approach, the alphas are derived from the CAPM.

[Insert Table 1 about here]

Importantly, to assure the robustness of our results, we apply the Fama-MacBeth regressions not only to raw returns, but also to risk-adjusted returns. The aim of this task is to see whether the *AMOM* and *AREV* signals predict not only raw payoffs, but also retain their forecasting ability after adjusting the payoffs with a multifactor model. Thus, we closely follow Avramov, Cheng, Schreiber, and Shemer (2015) and Avramov, Kaplanski, and Subrahmanyam (2018), and estimate the cross-sectional regression from the excess return adjusted previously with the three-factor model (2). To

estimate the model-adjusted payoffs, we replicate the approach of Jacobs (2015) and Zaremba (2016).

The results in Table 1 provide convincing evidence that the *AMOM* and *AREV* variables are highly significant return predictors in the cross-section, both in the simple regressions (Specifications [1] and [2]) and after controlling for each other (Specification [3]). Furthermore, they remain significant even after controlling for other variables—*BETA*, *MV*, *BM*, *IVOL*, *SKEW*, and *SEAS* (Specification [4])—and play an important role for both country (Panel A) and industry (Panel B) universes.⁵ The relationships hold not only when applied to raw excess returns (Panels A.1 and B.1), but also to model-adjusted returns (Panels A.2 and B.2). To sum up, alpha momentum and reversal are strong, robust, and independent predictors of future country and industry returns. Let us now see to what extent this phenomenon could be translated into profitable investment strategies.

6. Performance of Portfolios Formed on Past Alphas

We continue our analyses with the examination of portfolios from one-way sorts on past alphas. To this end, we rank all the country or industry indexes on *AMOM* and *AREV*, and weight them equally to obtain portfolios. We also form long-short portfolios, which go long in the indexes with the highest *AMOM* or *AREV* and simultaneously short the indexes with the lowest *AMOM* or *AREV* values. Finally, the analysis of portfolios from one-way sorts is supplemented by the test of monotonic relation (MR) introduced by Patton and Timmerman (2010). This is a simulation- test

⁵ The importance of these variables has been documented in the earlier literature. See for the market value: Keppler and Traub (1993), Keppler and Encinosa, (2011); for the EP ratio: Macedo (1995) and Kim (2012); for the beta: Frazzini and Pedersen (2014); for the idiosyncratic volatility: Bali and Cakici (2010) and Umutlu (2015); for skewness: Harvey (2000); for seasonality: Keloharju, Linnainmaa, and Nyberg (2016). Importantly, at this stage we do not consider the raw price-based momentum and reversal effects because these are discussed in detail later (see Tables 7–9).

aimed at detecting whether the returns on adjacent portfolios increase systematically along with the increase of the underlying variable used to sort the assets. The performance of the one-way sorted equal-weighted portfolios is reported in Table 2.

[Insert Table 2 about here]

In line with our expectations, the sorts on *AMOM* (Panel A of Table 2) are translated in a clear cross-sectional pattern in returns: the higher the *AMOM*, the higher the next month's payoff. The tendency is visible in both countries and industries and the monotonic pattern is confirmed with the formal MR test. The average returns on long-short portfolios amount to 0.84% and 1.15% for countries and industries, respectively. Notably, the differential returns remain positive and significant even after adjusting for the three-factor model of Fama and French (1993). The alpha for the country (industry) universe amounts to 0.92% (1.41%) with a corresponding *t*-statistic of 3.75 (5.98).

Panel B of Table 2 reports the performance of the portfolios formed on *AREV*. Although visibly lower (in absolute terms) than in the case of momentum, the long-short portfolios formed on *AREV* are also remarkable and significant, delivering a mean monthly return of -0.59% (-0.63%) for countries (industries). Again, the portfolios are characterized by a significant three-factor alpha of -0.52% (-0.49%). Also, the MR test strongly confirms the monotonicity in returns—both raw and three-factor model-adjusted—for countries and industries. Finally, although the raw profits on the differential portfolios are lower than those for the alpha momentum, the volatility of monthly returns decreased by more than one percentage point. Yet, the Sharpe ratios on the long-short portfolios

formed on *AMOM* are higher in absolute terms than for *AREV*, amounting to 0.58 (0.93) for country (industry) *AMOM* portfolios, and -0.48 (-0.66) for *AREV* portfolios.⁶

As a robustness check, Table 3 exhibits the performance of the portfolios from one-way sorts in an alternative setting, namely, within value-weighted portfolios. Some earlier studies (e.g., Zaremba, 2015, 2016) have argued that the country-level anomalies, including momentum in particular, are weaker for capitalization-weighted portfolios. This observation matches the findings in our sample. Although still positive and significant, the three-factor alphas on the long-short portfolios formed on momentum are noticeably lower as compared to equal-weighted strategies, amounting to 0.58% and 0.76% for the country and industry indexes, respectively. The monotonicity in the alpha momentum portfolios is still confirmed by the MR tests. Nonetheless, the performance of the long-short portfolios formed on *AREV* is much less impressive; the raw and risk-adjusted returns—both for countries and industries—are still negative, but very low in absolute terms and statistically insignificant. Also, the MR tests no longer point to a clear monotonicity in returns. This decline in performance may suggest that the alpha reversal effect is weaker in the large equity markets, which are generally more efficient and easier to arbitrage. This observation would be consistent with the behavioral story of overreaction-driven alpha momentum and reversal effects, as behavioral biases are less frequently observed in large, developed, and mature markets. We examine this issue in detail in a later part of this study.

[Insert Table 3 about here]

⁶ Also, we verify whether the results for alpha momentum and reversal hold after exclusion of the largest equity market—the United States—from the research sample. This operation leads to no qualitative changes in the results.

6.1. The Role of Additional Factors in Alpha Calculation

So far, we have relied on a very simple model to obtain alpha, namely CAPM. However, would the past alpha retain its predictive ability under alternative factor models as well? To verify this conjecture and check the robustness of the alpha-based strategies, we go beyond the simple CAPM and estimate the abnormal returns from three alternative models: (a) the three-factor model of Fama and French (1993), accounting also for the size and value effects in equities; (b) the four-factor model of Carhart (1997), which covers also the classic momentum effect; and (c) the three-factor model inspired by Asness, Frazzini, and Pedersen (2013), which includes three factors reflecting the market risk premium, the value effect, and the momentum effect. Furthermore, for robustness we check both the volatility-adjusted alphas as defined in Equation (5), and the sorts on a raw unadjusted alpha α_i as defined in Equations (1) to (4). The performance of long-short quintile portfolios—analogueous to the *T-B* portfolio in Table 2—formed under different models is reported in Table 4.

[Insert Table 4 about here]

First, the performance of alpha-based strategies is robust to alternative alpha estimation models. The two types of long-short portfolios—based on the alpha momentum and alpha reversal—yield positive and significant raw and three-factor model-adjusted returns in all the variants of alpha, regardless of the model employed. Furthermore, the overperformance is independent of the adjustment of the alpha for volatility (i.e., whether the alphas are volatility-adjusted or unadjusted) and of the asset universe (i.e., whether they are implemented within countries or industries). The profitability of the alpha-based strategies is pervasive.

The results in Table 4 also reveal a few additional insights. First, the strategies based on unadjusted alphas deliver slightly higher mean returns and abnormal returns than the portfolios based

on volatility-adjusted alphas, which is our default approach. Nevertheless, the lack of adjustment of the alphas also leads to the creation of marginally more volatile portfolios so that, overall, the Sharpe ratios are higher in the case of the volatility adjustment.

Second, the inclusion of additional factors in the factor models (i.e., *SMB*, *HML*, and *MOM*) does not lead to further improvement of the performance of the long-short portfolios. The mean returns and alphas are actually the highest for the strategies based on the return predictive signals derived from the simple CAPM framework. This suggests that most of the alpha momentum and reversal effects come from the market factor, and matches the findings of Blitz, Hanauer, and Vidojevic (2018) who, having investigated a similar concept—the idiosyncratic momentum in U.S. equities—also found the highest abnormal returns on the CAPM-derived residuals. In fact, this is also consistent with our overview of the explanatory power of the asset pricing models illustrated in Table A6 in the Online Appendix. Although the inclusion of *HML*, *SMB*, and *UMD* factors dramatically improves the explanatory power of the cross-sectional variation of returns, when it comes to the time-series variation, it seems that it is the market factor that plays the crucial role. In our brief tests, the CAPM explains on average 65.67% (75.87%) of the time-series variation, and adding the additional factors increases the average R^2 coefficient by only a few additional percentage points.

Nonetheless, when we focus on the alpha momentum (Panel A), the inclusion of further factors decreases not only profits, but also the volatility. This observation supports the view that basing the momentum on alphas as opposed to returns helps to reduce the style exposure, limiting the volatility. However, from the practitioner’s angle, the Sharpe ratios are still slightly higher for the CAPM-based portfolios than for the multifactor framework.

Interestingly, inclusion of the additional factors in the models does not lead to visible improvement of the risk-adjusted performance of the strategies. Although the significant profitability of the long-short portfolios holds for all models, it appears that simplicity prevails, and the multifactor alpha-based portfolio does not beat the simplest CAPM alpha. Apparently, the role of additional asset pricing factors is different at the country level than at the stock level.

6.2. The Influence of Trading Costs

Numerous papers have suggested that the profitability of strategies related to momentum might be noticeably impaired by trading costs (Korajczyk and Sadka 2004; Abbes, Boujelbène, and Bouri 2008; Andreu, Swinkels, and Tjong-A-Tjoe 2013). Indeed, Novy-Marx and Velikov (2016) and Chen and Velikov (2017) show that momentum strategies tend to be characterized by sizeable turnover, which translates into elevated transaction costs. On the other hand, the investment techniques based on long-run performance tend to display more stable portfolio composition and, in turn, a lower cost drag. Hence, we are interested in determining to what extent the alpha-based strategies are robust to trading costs.

To estimate the transaction costs, we track portfolio weights and apply a fixed percentage cost whenever a trade occurs. In other words, every time when a position is entered, exited, or rebalanced, we assume the investor pays a cost. For robustness, we assume four different values for the costs per one-way trade: 0.5%, 1.0%, 1.5%, and 2.0%. More formally, to estimate the total value of transaction costs, we multiply an assumed per-trade cost by the change of weights of the positions in the portfolio. In particular, we employ the equation (7):

$$c_t = c \sum_{i=1}^n |w_{i,t_{b+1}} - w_{i,t_e}|, \quad (7)$$

where c_t is the total transaction cost of a portfolio of n country or industry indexes in month t , w_{i,t_e} is the actual weight at the end of month t , $w_{i,t_{b+1}}$ is the target weight at the beginning of month $t+1$, and c is the cost proxy amounting to 0.5%, 1.0%, 1.5%, or 2.0%. The cost-adjusted performances of the portfolios formed on past alphas are presented in Table 5. For brevity, we limit our attention to the equal-weighted portfolios only, which, as a rule, tend to have higher trading costs.

[Insert Table 5 about here]

Both alpha-based country-selection strategies—alpha momentum and alpha reversal—remain significantly profitable, even after applying a hefty and conservative value of 1.0% to estimate the trading costs. The implementation of the portfolios within the industry universe leads to even higher resilience, with profits remaining significant even up to the 2.0% (1.5%) cost level applied to the alpha momentum (reversal) portfolios.

6.3. Alternative Holding Periods

We are also interested in the performance of the alpha momentum and reversal strategies under alternative—longer—holding periods. To this end, we investigate 12 different holding periods from 1 to 12 months. The strategies are subject to monthly rebalancing and a portfolio with an n -month holding period aggregates overlapping equal-weighted strategies based on signals from the last n months. Table 6 reports the performance of the long-short quintile portfolios—analogueous to Table 2—formed under assumptions of alternative holding periods from 1 to 12 months.

[Insert Table 6 about here]

The alpha-based strategies retain their profitability even when the holding period is visibly extended. For country (industry) momentum portfolios, the alphas remain positive and significant

even when the indexes are held to 10 (12) months. Only later are they indistinguishable from zero. For the reversal, the performance actually improves when the holding period is slightly longer. Under different specifications, the absolute payoffs are the highest around the 3–5 month holding period, displaying also stronger significance. More importantly, this approach would also result in lower trading costs, leading to further improvement of the post-cost performance. To sum up, the alpha momentum and reversal strategies are robust to the application of much longer holding periods, even sometimes leading to—as in the case of reversal strategy—improvement in performance.

6.4. Controlling for Price Momentum and Reversal Strategies

The alpha momentum and reversal strategies appear to perform very well, delivering sizeable abnormal returns and Sharpe ratios. However, we are particularly interested in whether their raw and risk-adjusted performances remain significant after controlling for the roles of the price-based momentum and reversal effects. This exercise is particularly important from the asset pricing perspective. We would like to know whether the alpha-based momentum and reversal phenomena are separate asset pricing factors, or if they are just manifestations of the price momentum and reversal. Or, maybe, it is the price momentum and reversal that are manifestations of their alpha-based counterparts. In other words, can the regular return-based momentum (reversal) effect explain the alpha momentum (reversal) phenomenon? Or is it the alpha momentum (reversal) that drives the price momentum (reversal)?

To check this, we apply three different and independent tests: (a) Fama-Macbeth regressions controlling for past returns, (b) time-series spanning tests, and (c) evaluation of portfolios from two-way dependent sorts on the alpha-based and return-based strategies. Each of these tests has its

strengths and weaknesses, so conducting them together helps us to draw unbiased conclusions.⁷ Also, our examinations follow the spirit of Fama (2015), who argues that cross-sectional and time-series tests should be performed jointly because they provide unique perspectives that complement each other.

We start with the Fama-MacBeth (1973) regressions, the results of which are reported in Table 7. In this approach, our baseline regression model is represented by Equation (6), and we also include a set of additional variables: *MOM* and *REV*. In particular, we are interested in seeing whether *AMOM* (*AREV*) keeps its cross-sectional predictive ability for future returns after controlling for other widely used determinants of return, including *MOM* (*REV*).

[Insert Table 7 about here]

The coefficient estimate for the *AMOM* variable is positive and significant across all the specifications (i.e., in regressions applied to both raw and risk-adjusted returns). On the other hand, the slope estimate for the *MOM* variable, representing the price momentum, turns insignificant for nearly all specifications that include *AMOM*, suggesting that the price momentum is fully captured by the alpha momentum. These observations support the theory that the alpha momentum is likely an underlying driver of the price momentum.

The results for the *AREV* signal are a bit less clear. On the one hand, the regular return reversal effect is insignificant across all the specifications – once we control for the alpha reversal, the price reversal is no longer significant. Again, this supports the view that the alpha-based measures constitute the true drivers of the return reversal. On the other hand, the alpha reversal is significant

⁷ We are aware that each of the methods has its limitations. The time-series spanning tests depend on the arbitrary portfolio construction methods, which may influence the final results. The two-way dependent sorts may suffer from a small sample bias: splitting 51 stock market indexes into nine portfolios results in portfolios with few components. Finally, the Fama-MacBeth regressions may experience problems stemming from collinearity of some of the variables. Therefore, we apply three independent tests, assuming that consistent results would be evidence supporting or refuting the view that alpha-based momentum or reversal phenomena are independent asset pricing factors.

only for industries (both in raw and risk-adjusted returns). For country indexes, both return and alpha reversals—when considered jointly—prove insignificant.

One of the weaknesses of cross-sectional regression, which may be responsible for some of the insignificant outcomes in Table 7, is its inability to deal with highly correlated variables. In other words, if a pair of variables in the regression is correlated in the cross-section, the results might be biased and should be interpreted with caution. A quick review of the results reported in Table A7 of the Online Appendix unequivocally indicates that the alpha momentum and reversal are strongly correlated with their return-based counterparts. Thus, we supplement our results with the second examination: the time-series spanning test.

When implementing the time-series spanning tests, we closely follow the approach of Blitz, Hanauer, and Vidojevic (2018). The mean-variance spanning test, originally proposed by Huberman and Kandel (1987), helps to answer the question of whether the alpha-based strategies add any incremental value from the perspective of an investor who holds a diversified portfolio, such as a market portfolio or the factor portfolios from the Fama-French (1993) model. Specifically, the idea of the mean-variance spanning test in our case is to show whether the alpha momentum (reversal) portfolio lies outside the mean-variance frontier spanned by asset pricing factor returns, including the momentum (reversal) factor. In practice, we run a time-series regression of the long-short alpha momentum and reversal portfolios on the existing factor returns from asset pricing models. Significant intercepts in the regression indicate that the existing set of factors is not able to explain the abnormal returns on the tested portfolio which, in turn, expands the mean-variance frontier.

Closely following the framework of Blitz, Hanauer, and Vidojevic (2018), we form all the factor portfolios in the test in a consistent way, closely replicating the technique of Fama and French (1993). Thus, the factor portfolios are constructed using independent sorts on the country or industry

indexes in two size-related and three alpha momentum or reversal-related groups. The indexes in the respective groups are then value-weighted to obtain the component portfolios. The alpha momentum factor return is then calculated as the average return on the two high-*AMOM* portfolios (large and small markets) minus the average return on the two low-alpha momentum portfolios. Regarding the alpha reversal (and return reversal as well), we introduce a small change to assure better comparability of results. In particular, the alpha reversal factor return is the average return on the two low-*AMOM* portfolios (large and small markets) minus the average return on the two high-*AREV* portfolios. Thanks to this operation, the long-short alpha factor produces on average *positive*, not *negative* returns.

The mean-variance spanning test provides two interesting insights. First, according to their original interpretation, the Fama-French model factors are proxies for risks faced by equity investors. If they play a significant role or explain the tested portfolio's abnormal returns, then it would provide evidence that the portfolio payoffs are at least partly risk-driven. Otherwise, lack of explanatory power would support the view of behavioral mispricing.

Second, we also include in the regressions the factor portfolios reflecting the price momentum and reversal effects, following the same procedures as the alpha-based counterparts. Thus, we want to see whether the alpha-based approach offers incremental value even to these diversified investors who already implement the return-based momentum and reversal strategies.

The results of the factor spanning tests are presented in Table 8. The Fama-French factors, namely *MKT*, *SMB*, and *HML*, are not able to explain the alpha momentum or alpha reversal performance. Admittedly, the examined portfolios do exhibit some exposure to the *HML* factor, and the intercepts remain high and significant. The abnormal returns on country portfolios formed on *AMOM* and *AREV* amount to 0.95% (t -stat = 4.39) and 0.43% (t -stat = 2.08), respectively. For

industries, the analogous numbers equal 0.99% (t -stat = 5.24) and 0.21% (t -stat = 1.71). This supports our initial view that the relationship between future returns and past alphas is driven by behavioral factors and investor sentiment rather than risk factors.

[Insert Table 8 about here]

Second, we observe the abnormal returns on the alpha momentum strategies after controlling for the price momentum returns. They remain positive and significant in all the universes and settings. On the other hand, the price momentum portfolios' payoffs are fully spanned by the alpha momentum portfolios, once again supporting our view of the dominance of the *AMOM* framework and matching the findings in Table 7. However, the performance of the alpha reversal portfolios is, again, slightly more complicated. On the one hand, the *AREV* factors fully explain the returns on the *REV* factors. None of these portfolios—either in the country or industry universes—produces significant abnormal returns once we control for *REV*. In other words, alpha reversal fully covers the return reversal. However, when we control for *REV*, the *AREV* returns are only significant within country portfolios; for industries, the alphas are indistinguishable from zero. Summing up, the results for the alpha reversal effect are once again mixed, providing only partial support for the hypothesis that price reversal is only a manifestation of the alpha reversal. What is worse, the partial evidence is found in a different universe: not for industries, as in the Fama-MacBeth regressions disclosed in Table 7, but for countries. To shed some further light on this problem, we continue with our third test, namely, the evaluation of the portfolios from two-way sorts.

We implement the third test to see whether the alpha-based strategies remain significant after controlling for their return-based counterparts; the test examines the performance of portfolios from double dependent sorts. In this approach, we conduct two-way sorts of indexes in our sample. In the first pass, we sort the indexes based on *MOM* (*REV*) into three portfolios. In the second pass, we

continue with sorting the indexes within each *MOM* (*REV*) tertile into three portfolios based on *AMOM* (*AREV*). These two-way sorts result in nine portfolios and allow us to examine the relationship between *AMOM* (*AREV*) and future index returns after controlling for the individual momentum (reversal) strategy. Finally, having completed this task, we also reverse the procedure and begin with the sorts on *AMOM* and *AREV* in the first pass, and continue with the sorts on *MOM* and *REV* in the second pass. In this setting, we investigate whether the price momentum and reversal effects hold after controlling for their alpha counterparts.⁸

Table 9 shows the three-factor model alphas for the portfolios from dependent two-way sorts on alpha and return signals.⁹ Panel A of Table 9, reporting the outcomes of sorts on past alphas within return tertiles, confirms once again the reliability of the alpha momentum effect. In the vast majority of cases, the alpha momentum long-short portfolios (High-Low) within *MOM* tertiles deliver positive significant abnormal returns, indicating that there is a positive relationship between *AMOM* and expected index returns for portfolios with similar levels of *MOM*. The lower section of Panel A focuses on the *AREV* sorts within *REV* tertiles. As in Tables 7 and 8, the abnormal returns on the long-short *AREV* portfolios are only partially significant. Specifically, they significantly depart from zero in three out of six cases, predominantly among the indexes displaying high long-term past returns. For the indexes with low long-term returns, the payoffs on the long-short *AREV* portfolios are only insignificantly negative.

[Insert Table 9 about here]

⁸ One of the limitations of the use of two-way sorted portfolios for countries is that the average portfolio includes less than 6 indexes on average. For industries, the total sample includes 887 assets, so the number of indexes in a portfolio is not a concern.

⁹ The raw excess returns on these portfolios are reported in Table A8 of the Online Appendix. The examination of raw returns leads to identical conclusions.

Panel B of Table 9, reporting the performance of portfolios from sorts on returns within alpha tertiles, brings a host of further interesting insights. With the exception of only one long-short portfolio (for high alphas on industry indexes), the price momentum effect does not exist in any of the *AMOM* tertiles. Similarly, the long-run reversal is not present in any of the *AREV* tertiles. In other words, after accounting for *AREV* and *AMOM*, the regular price momentum and reversal phenomenon almost completely disappear.

Our conclusions from Tables 7 to 9 could be summarized as follows. The alpha momentum effect entirely subsumes the price momentum pattern: it remains significant after controlling for the price momentum, but the price momentum ceases to exist once we control for the alpha momentum. This supports the view that the price momentum is simply a manifestation of the alpha momentum. Regarding the reversal, our evidence in support of a parallel view is only partial. The return reversal is insignificant in all cases once we control for the alpha reversal. On the other hand, the alpha reversal subsumes the price reversal only in selected sample segments such as portfolios with past long-term returns. It seems that the *AREV* and *REV* variables are closely linked and the superiority of *AREV* over *REV* is relatively weaker than the superiority of *AMOM* over *MOM*.

6.5. The Past Alpha and Limits to Arbitrage

So far, we have provided evidence that the alpha momentum and reversal strategies are strong and robust predictors of future industry and country returns in the cross-section. However, could they be further improved? The behavioral view of equity anomalies asserts that they result from investor irrationality that cannot be easily arbitrated away. If the momentum and reversal effects observed in indexes are behavioral phenomena resulting from overreaction, they should be particularly pronounced in the market segments characterized by higher limits to arbitrage. Recent

studies empirically examined the influence of cross-sectionally varying limits to arbitrage on various market anomalies, including the book-to-market effect (Ali, Hwang, and Trombley 2003), momentum (Nagel 2005), accrual anomaly (Mashruwala, Rajgopal, and Shevlin 2006), asset growth anomaly (Lam and John Wei 2011), or low-risk anomaly (Gu, Kang, and Xu 2018). To test whether the alpha momentum and reversal strategies are stronger in market segments characterized by higher limits to arbitrage, we follow the approach of Gu, Kang, and Xu (2018), examining the performance of portfolios from double-sorts on several stock-return predictors and proxies of limits to arbitrage. For robustness, we use three different proxies motivated by the previous literature, assuming that the equity anomalies are more pronounced for stocks with high idiosyncratic risk (Jiang, Lee, and Zhang 2015), low R^2 (Hou, Xiong, and Peng 2006), and low capitalization (Hong, Lim, and Stein 2000; Zhang 2006). Specifically, each month we sort the markets on alpha-based strategies and a) the idiosyncratic volatility from the CAPM based on a rolling 60-month window, (b) the R^2 coefficient from the three-factor model (2) based on a rolling 60-month window, and (c) total stock market capitalization in month $t-1$, to obtain nine portfolios from dependent two-way sorts, analogously to Section 6.4. Table 10 demonstrates the three-factor model alphas from the double-sorts on limits to arbitrage.¹⁰

[Insert Table 10 about here]

As expected, the performance of the alpha momentum and reversal strategies is indeed better in the markets with higher limits to arbitrage. Focusing first on Panel A.1 of Table 10, the three-factor model alpha on the long-short portfolio from sorts on *AMOM* within the small country (industry) tertile equals 0.87% (1.63%), whereas within the large size tertile it equals 0.25%

¹⁰ The average excess returns on these portfolios are reported in Table A9 of the Online Appendix. The raw returns point to no major qualitative differences in the results.

(0.70%). Regarding the levels of idiosyncratic risk (Panel A.2 of Table 10), the long-short alpha momentum portfolios produce an alpha of 0.76% (1.24%) in the high-idiosyncratic volatility countries (industries) and only 0.27% (0.63%) when the idiosyncratic volatility is low. Finally, the R^2 coefficient plays a similar role (Panel A.3 of Table 10). The long-short portfolios deliver an alpha of 0.94% (1.53%) in the low R^2 country (industry) indexes and 0.65% (0.79%) in the high R^2 country (industry) indexes. In summary, the higher the limits to arbitrage in the cross-section, the better is the performance. This evidence supports our view that the return cycles related to past alpha momentum are behaviorally driven phenomena.

The effects of the limits to arbitrage on the alpha reversal effect are even more striking. The effect of *AREV* on future returns is present only in low-capitalization, high-idiosyncratic volatility, and low- R^2 markets. In the opposite segments (i.e., across the large countries characterized by low *IVOL* and high R^2), the alpha reversal effect simply does not exist. For instance, in the tertile of large countries (industries), the three-factor model alpha amounts to an insignificant -0.14% (-0.11%) per month. This observation also explains why the alpha reversal effect works so poorly in capitalization-weighted portfolios as documented in the previous sections.

Supplementary to the results in Table 10, we also test the profitability of *AMOM* and *AREV* signals within developed and emerging markets. Our reasoning in this case is similar to that for the tests reported in Table 10: if alpha momentum and reversal phenomena are driven by overreaction, they should be less pronounced in developed markets than in emerging ones. In emerging economies, the limits to arbitrage are stronger, driven by large short-sale constraints, limited leverage availability, lower liquidity, or even more challenging cross-country mobility

constraints. In consequence, for example, Zaremba and Szczygielski (2018) document that certain cross-country value strategies are profitable only in emerging markets.

To check the effect of market development, we split the sample into emerging and developed markets following the Datastream classification, and test the performance of *AMOM* and *AREV* within tertile portfolios of these two subsamples. The results are reported in Table 11. Indeed, consistent with our hypothesis, raw and risk-adjusted returns are usually higher in absolute terms in the emerging market subsample. The effect is particularly pronounced for the reversal portfolios. In this case, the alphas on the country (industry) portfolios equal -0.28% (-0.21%) in developed markets and -0.70% (-0.40%) in the emerging ones. In other words, the phenomenon is twice as strong in the less developed equity universes.

[Insert Table 11 here]

6.6. Performance within Subperiods

Value and reversal strategies tend to behave differently in different subperiods (Asness, Moskowitz, and Pedersen, 2013). Hence, for robustness, we examine a range of different subperiods motivated by the finance literature. In particular, we focus on the insights from behavioral finance that have implications for our overreaction hypothesis of the alpha momentum-reversal cycle. First, recent research suggests that there have been recent declines in the profitability of many equity anomalies (McLean and Pontiff 2016). Investor learning and improvement of market efficiency lead to a decline in behavioral mispricing. Importantly, the momentum effect is no exception (Bhattacharya, Kumar, and Sonaer 2017; Dolvin and Foltice 2016). Thus, we split the full time-series into two roughly equal subsamples: January 1972 to March 1995, and April 1995 to May 2018.

Second, as we have indicated, there tends to be a January effect for the momentum and reversal anomalies (Haug and Hirshey 2006; George and Hwang 2007). Haug and Hirshey (2006) and Brown (2016), among others, show that the momentum strategy produces disappointing performance in January, whereas George and Hwang (2007) argue that the long-run reversal effect is the strongest in this month. Hence, we explore the profitability in January and non-January months.

Third, Cooper, Gutierrez, and Hameed (2004) hypothesize that overreaction should be particularly high (weak) following bull (bear) markets, resulting in more pronounced momentum and reversal payoffs. Drawing from this study, we investigate the performance of the alpha momentum and reversal portfolios in the months right after bear and bull markets. For month t , we assume three alternative definitions of the bear (bull) markets—we define a market as a bull (bear) market if the average return through the prior 24, 36, and 48-month window is positive (negative).

Fourth, Jacobs (2015) argues that the performance of anomalies tends to be affected by time-varying measures of limits to arbitrage, such as liquidity (Avramov, Cheng, and Hameed, 2016) or volatility (Wang and Xu, 2015). Inspired by these studies, we examine the portfolios formed on past alphas within periods of high and low limits to arbitrage. We use three different measures of limits to arbitrage which were previously employed in studies by Jacobs (2015) and Zaremba (2016): (a) the *VIX* volatility index (*VIX*); (b) BAA credit spread (*Credit*) (c) *TED* spread (*TED*), closely following the methods of calculation and implementation of Zaremba (2016).¹¹ For each of these measures, we compute the median value. Subsequently, we investigate the returns on the alpha momentum and reversal strategies separately in the subperiods when the *VIX*, *Credit*, *Term*, and *TED* are below and above the median at the end of month $t-1$.

¹¹ All data are retrieved from Bloomberg.

Fifth, Stambaugh, Yu, and Yuan (2012) argue that the profitability of equity anomalies is influenced by time-varying investor sentiment. Therefore, we examine the payoffs on the alpha-based portfolios within periods of high and low investor sentiment. For robustness, we use three different measures: the Baker and Wurgler (2006) sentiment index (*BW*), the Index of Consumer Expectations calculated by the University of Michigan (*MICE*), and the European Commission's Economic Sentiment Indicator (*ECES*).¹² Again, we replicate the same procedures to determine the months of below-median and above-median investors' sentiment.

Table 12 displays the mean returns for the long-short equal-weighted quintile portfolios of countries and industries (i.e., the basic *T-B* portfolios from Table 2) from sorts on the alpha momentum (Panel A) and alpha reversal (Panel B).¹³

[Insert Table 10 about here]

The results display empirical evidence consistent with our behavioral explanation, particularly in the case of the alpha reversal phenomenon. The performance of the momentum strategy is very stable—the portfolio delivers significant average returns in majority of the subperiods. However, the average returns are insignificant in January and significant only in other months. More importantly, the alpha momentum effect is visibly stronger following bull markets than bear markets, in line with the implications of the overreaction hypothesis by Cooper, Gutierrez, and Hameed (2004). And, for the country portfolios, the mean alpha momentum returns do not

¹² The data on *BW* are obtained from Jeffrey Wurgler: <http://people.stern.nyu.edu/jwurgler/>. The data on *CBCC* and *ECES* are retrieved from Bloomberg. Additional information on *CBCC* can be found at <https://www.conference-board.org/data/consumerconfidence.cfm>; for *ECES*: https://ec.europa.eu/info/index_en.

¹³ The three-factor model alphas on these portfolios are reported in Table A10 in the Online Appendix. The results are qualitatively similar, but less clear. The reason is that the estimated alpha depends on factor realizations within the subperiods, which in turn—if also behaviorally driven—might be equally affected by the time-varying variables considered in this test.

significantly differ from zero, no matter how long the bull market measurement period is: 24, 46, or 48 months.

The times-series variability of the reversal effect is even more pronounced. The effect visibly diminishes in the second half of our sample, consistent with the learning hypothesis of McLean and Pontiff (2015). It is also markedly stronger in the periods of high limits to arbitrage, defined as high *VIX*, *Credit*, or *TED*. For country portfolios it is actually insignificant in the other subperiods. The effect is also stronger in times of high investor sentiment, displaying larger absolute returns. Also, in line with the view of George and Hwang (2007), the effect is stronger in January. Finally, similar to momentum, the alpha reversal is remarkably more pronounced and significant following bull markets as compared to following bear markets. All these observations support the hypothesis of the behavioral roots of the alpha reversal anomaly.

7. Concluding Remarks

This study examines the predictive abilities of past alphas for returns on country and industry indexes. We show that past short-term alphas (6–12 months) positively forecast future returns, whereas long-term alphas (6–120 months, lagged 12 months) exhibit a negative correlation. The performance of the strategies—and the alpha momentum in particular—is robust to many considerations including the use of alternative factor models to estimate alpha, the influence of trading costs, and alternative portfolio weighting methods. The results also hold for subsample and subperiod analyses, and even after controlling for other well-known predictive variables.

Importantly, the alpha momentum effect fully subsumes the price momentum effect: it fully explains the price momentum profits, but the alpha momentum is not explained by the price

momentum. For the long-run reversal pattern, this superiority is less dramatic, but the price reversal effect still becomes insignificant once we control for the alpha reversal.

We also focus on the potential sources of the alpha momentum and reversal effects. Our results are mostly aligned with the behavioral explanations of the two phenomena, accentuating the underreaction and overreaction cycles. Consistent with this, the alpha momentum and reversal are particularly strong in the market segments characterized by high limits on arbitrage, i.e., within small markets of high idiosyncratic volatility and low correlation with the global equity market. Also, the returns are particularly elevated following bull markets, supporting the behavioral mispricing explanation.

The findings of this study not only provide new insights for understanding anomalies at the country- and industry-level, but also reveal practical implications for global portfolio management. Our paper offers a novel tool that could be directly used for international asset allocation across countries and industries.

Future studies can extend the use of alpha-based strategies to other asset classes. Could a similar approach be applied to factor premia in, for instance, commodities or corporate bonds? It would be interesting to shed some light on these issues.

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Figures

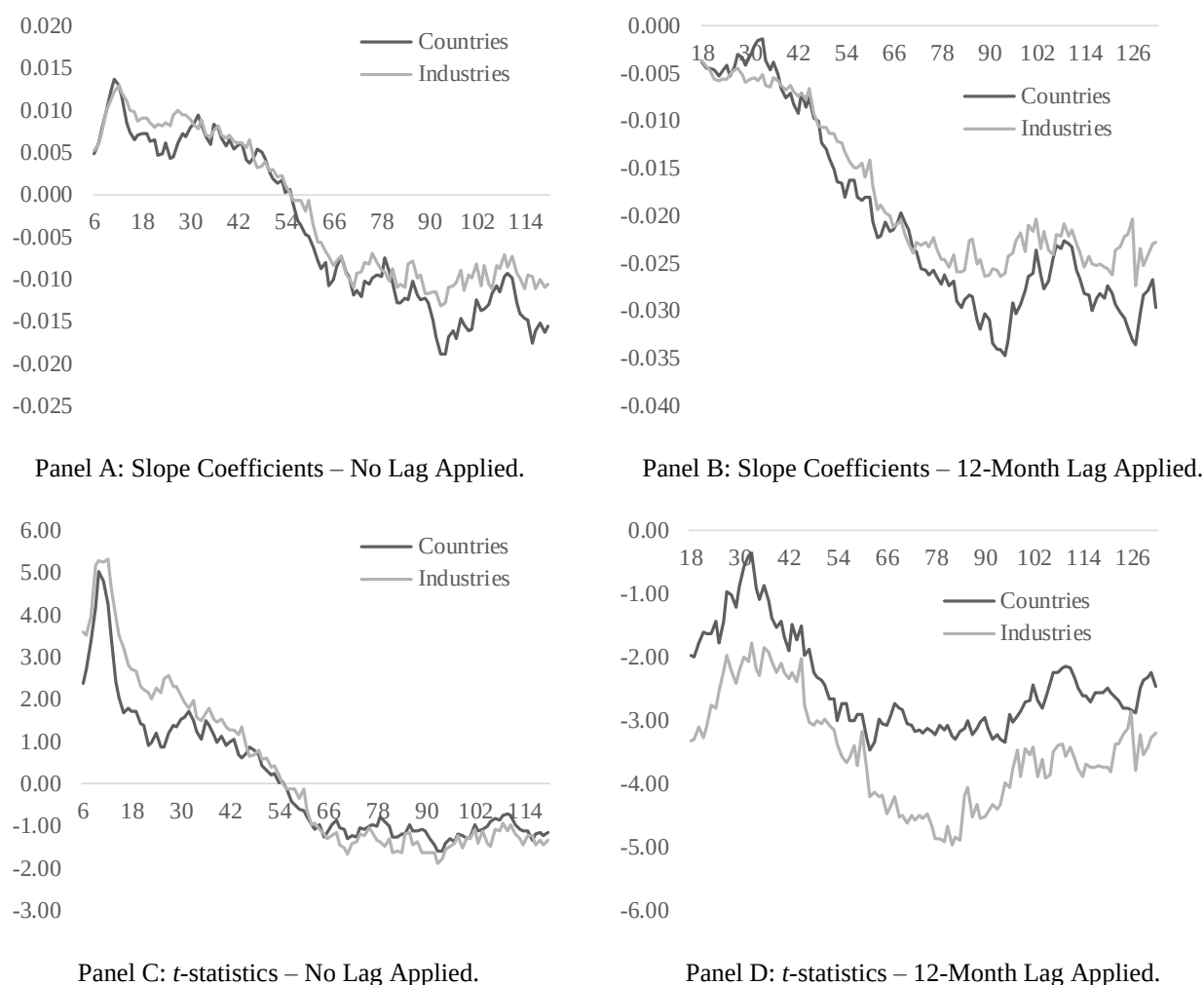
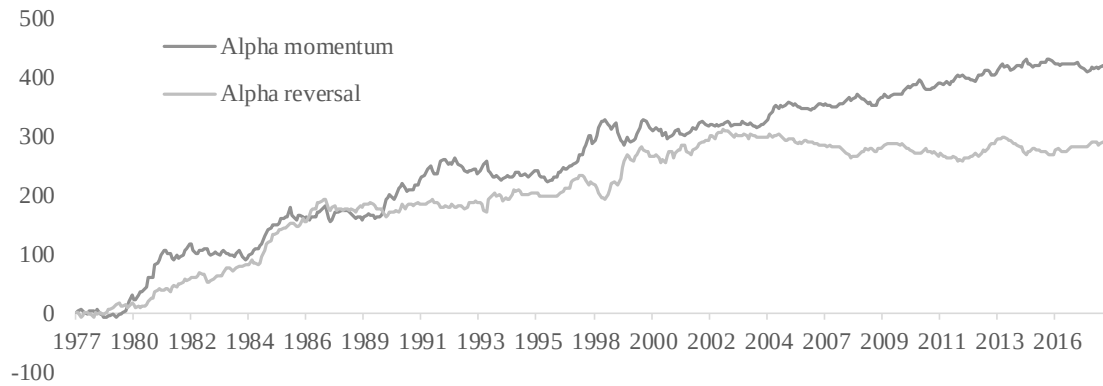


Figure 1. Slope Coefficients and t -Statistics from Fama-MacBeth Regressions on Past Alphas

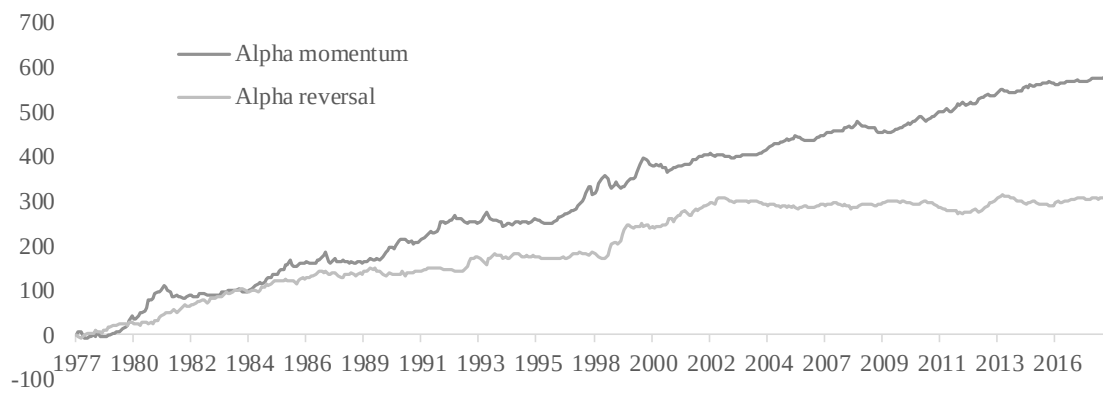
This figure presents the slope coefficients (Panels A and B) with the corresponding t -statistics (Panels C and D) from univariate Fama and MacBeth (1973) regressions applied to returns on country and industry portfolios according to Equation (1):

$$R_{i,t} = \beta_0 + \beta_1 \alpha_{i,t-k} + \varepsilon_{i,t+1},$$

where $R_{i,t}$ is the return on the portfolio i in month t , $\alpha_{i,t-k}$ is the monthly volatility-adjusted alpha from the CAPM on the portfolio i estimated through the trailing k months, and β_0 and β_1 are regression parameters. The regressions use monthly data from the period January 1973 to October 2017. Panels A and C present parameters estimated based on the full k -month period, whereas Panels B and D present the parameters estimated with the most recent 12 months discarded from the estimation period.



Panel A: Country Indexes



Panel B: Industry Indexes

Figure 2. Cumulative Returns on Long-Short Alpha Momentum and Alpha Reversal Portfolios

Note. This figure presents the cumulative returns on long-short equal-weighted quintile portfolios formed on the *alpha momentum* (AMOM), i.e., the volatility-adjusted CAPM alpha estimated through months $t-12$ to $t-1$ and the *alpha reversal* (AREV), i.e., the volatility-adjusted the CAPM alpha estimated through months $t-60$ to $t-13$. The portfolios go long (short) the indexes with the highest (lowest) AMOM and the lowest (highest) AREV. The returns are expressed as percentages and cumulative returns are estimated additively. Panels A and B report results for the country and industry index universes, respectively.

Tables

Table 1

Results of Cross-Sectional Fama-MacBeth Regressions on Past Alpha

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Panel A: Countries								
Panel A.1: Raw excess returns				Panel A.2: Risk-adjusted returns				
AMOM	1.29*** (4.26)		1.18*** (3.70)	0.99*** (2.92)	1.04*** (3.67)		0.97*** (3.32)	0.81** (2.39)
AREV		-1.80*** (-2.90)	-1.53*** (-2.74)	-1.44** (-2.00)		-1.87*** (-3.18)	-1.61*** (-3.05)	-1.57** (-2.16)
BETA				-0.05 (-0.19)				-0.15 (-0.50)
MV				0.04 (0.80)				0.04 (0.75)
EP				8.82*** (3.50)				6.66*** (2.84)
IVOL				6.14 (1.27)				4.02 (0.84)
SKEW				-0.02 (-0.17)				-0.02 (-0.19)
SEAS				3.26 (0.75)				0.75 (0.18)
R ²	8.28	6.25	14.02	44.75	7.76	6.22	13.40	40.78
Panel B: Industries								
Panel B.1: Raw excess returns				Panel B.2: Risk-adjusted returns				
AMOM	1.30*** (5.32)		1.30*** (5.31)	1.34*** (5.53)	1.04*** (4.53)		1.05*** (4.08)	1.11*** (4.58)
AREV		-1.42*** (-3.19)	-1.47*** (-3.58)	-1.20*** (-3.09)		-2.07*** (-5.36)	-2.13*** (-5.84)	-1.77*** (-4.73)
BETA				0.06 (0.37)				-0.03 (-0.26)
EP				-0.10*** (-4.08)				-0.07*** (-3.37)
BM				2.82*** (3.71)				2.57*** (4.01)
IVOL				3.70* (1.80)				0.23 (0.12)
SKEW				-0.04 (-0.67)				-0.03 (-0.52)
SEAS				4.38*** (2.83)				1.69 (1.08)
R ²	2.60	1.79	4.57	14.60	2.26	1.64	4.04	11.62

Note. This table reports the slope coefficients and the corresponding t-statistics from the monthly Fama-MacBeth cross-sectional regressions in the following form:

$$R_{i,t} = \beta_{0,t} + \sum_{j=1}^J \beta_{j,t} K_{i,t-1} + \varepsilon_{i,t},$$

where $R_{i,t}$ is the excess return (Panels A.1 and B.1) or three-factor model-adjusted returns (Panels A.2 and B.2) on index i in month t ; $\beta_{0,t}$ and $\beta_{j,t}$ are regression parameters; and $K_{i,t-1}$ are various return predictors: volatility-adjusted alpha from the CAPM estimated through the months $t-12$ to $t-1$ (AMOM); volatility-adjusted alpha from the CAPM estimated through the months $t-60$ to $t-13$ (AREV); stock market beta (BETA) and idiosyncratic volatility (IVOL) estimated with the CAPM through the months $t-60$ to $t-1$; log-market value at month $t-1$ (MV); earnings-to-price ratio (EP); skewness of the return distribution estimated through the months $t-24$ to $t-1$ (SKEW), and the cross-sectional seasonality, i.e., average return in the same calendar month through the months $t-240$ to $t-1$, as available (SEAS). R^2 is the average cross-sectional coefficient of determination (expressed in %). The values in parentheses are Newey-West (1987) adjusted t -statistics. The stars *, **, and *** denote values reliably departing from 0 at the 10%, 5%, and 1% levels, respectively. The stars ***, **, and * indicate statistical significance the 1%, 5%, and 10% levels, respectively.

Table 2

Performance of the Equal-Weighted Portfolios from One-Way Sorts on Past Alpha

	Countries							Industries						
	Bottom	2	3	4	Top	T-B	MR	Bottom	2	3	4	Top	T-B	MR
<i>Panel A: Alpha momentum: sorts on performance through months $t-12$ to $t-1$</i>														
R	0.39 (1.35)	0.51* (1.95)	0.65** (2.55)	1.01*** (3.98)	1.22*** (5.56)	0.84*** (3.75)	0.01***	0.48* (1.86)	0.53** (2.24)	0.89*** (4.07)	1.20*** (5.60)	1.63*** (7.72)	1.15*** (5.97)	0.92***
Vol	5.49	5.43	5.32	5.58	5.47	5.03		5.31	4.87	4.71	4.84	4.99	4.28	
SR	0.25	0.33	0.42	0.63	0.77	0.58		0.31	0.38	0.65	0.86	1.13	0.93	
α	-0.37** (-2.37)	-0.27** (-2.29)	-0.05 (-0.42)	0.28** (2.51)	0.55*** (3.48)	0.92*** (3.75)	0.01***	-0.45*** (-3.35)	-0.31*** (-4.25)	0.10 (1.53)	0.47*** (5.60)	0.95*** (6.83)	1.41*** (5.98)	0.00***
MKT	0.94*** (25.17)	0.99*** (34.55)	1.04*** (36.91)	1.11*** (32.81)	0.94*** (19.25)	0.01 (0.11)		1.00*** (26.70)	1.02*** (50.81)	0.99*** (64.51)	1.01*** (44.89)	0.93*** (23.61)	-0.06 (-0.91)	
SMB	0.38*** (7.17)	0.44*** (10.16)	0.32*** (5.64)	0.39*** (6.61)	0.56*** (6.05)	0.18 (1.55)		0.72*** (10.02)	0.65*** (16.49)	0.73*** (22.58)	0.71*** (15.91)	0.88*** (11.58)	0.16 (1.16)	
HML	0.35*** (6.78)	0.30*** (6.51)	0.13*** (3.04)	0.04 (1.01)	0.01 (0.16)	-0.34*** (-3.84)		0.36*** (4.79)	0.21*** (5.42)	0.10*** (2.78)	-0.01 (-0.25)	-0.14** (-2.43)	-0.50*** (-5.05)	
<i>Panel B: Alpha reversal: sorts on performance through months $t-60$ to $t-13$</i>														
R	1.05*** (3.99)	0.96*** (3.59)	0.85*** (3.12)	0.78*** (2.98)	0.45* (1.92)	-0.59*** (-2.81)	0.05***	1.22*** (4.65)	1.02*** (4.19)	1.04*** (4.45)	0.89*** (4.16)	0.59*** (2.71)	-0.63*** (-3.96)	0.00***
Vol	5.27	5.40	5.45	5.44	5.10	4.29		5.09	4.81	4.79	4.63	4.76	3.29	
SR	0.69	0.62	0.54	0.50	0.31	-0.48		0.83	0.73	0.75	0.67	0.43	-0.66	
α	0.28* (1.90)	0.13 (1.23)	0.04 (0.34)	-0.01 (-0.10)	-0.24* (-1.70)	-0.52** (-2.33)	0.35***	0.32*** (2.83)	0.14* (1.92)	0.20** (2.36)	0.07 (0.89)	-0.17* (-1.86)	-0.49*** (-2.88)	9.94*
MKT	0.98*** (26.39)	1.06*** (36.25)	1.05*** (37.36)	1.08*** (36.04)	0.95*** (22.45)	-0.03 (-0.41)		1.01*** (33.56)	1.03*** (58.99)	1.03*** (54.10)	0.99*** (49.26)	1.00*** (39.42)	-0.01 (-0.29)	
SMB	0.44*** (7.84)	0.45*** (9.79)	0.37*** (8.28)	0.44*** (8.50)	0.38*** (5.96)	-0.06 (-0.59)		0.73*** (12.17)	0.64*** (16.66)	0.62*** (14.07)	0.62*** (17.10)	0.65*** (13.67)	-0.08 (-0.98)	
HML	0.21*** (4.63)	0.26*** (5.89)	0.29*** (5.55)	0.11** (2.34)	0.06 (1.44)	-0.15** (-2.21)		0.18*** (3.69)	0.18*** (4.34)	0.12** (2.20)	0.10** (2.11)	-0.04 (-0.89)	-0.22*** (-3.12)	

Note. The table reports the performance of equal-weighted quintile portfolios of country (left-hand side) and industry (right-hand side) equity indexes from sorts on: the *alpha momentum*, i.e., the volatility-adjusted CAPM alpha estimated through months $t-12$ to $t-1$ (Panel A) and the *alpha reversal*, i.e., the volatility-adjusted CAPM alpha estimated through months $t-60$ to $t-13$ (Panel B). The *T-B* is the long-short portfolio going long in the *Top* and short in the *Bottom* portfolios. *R* is the mean monthly excess return; *Vol* is the standard deviation of monthly returns; and *SR* is the annualized Sharpe ratio. α is the alpha from the Fama-French (1993) three-factor model, and *MKT*, *SMB*, and *HML* are the measures of exposure of the market portfolio, small minus big, and high minus low factors, respectively. *MR* is the *p*-value from the Patton and Timmerman test of monotonic relationship. *R*, *Vol*, α , and *MR* are presented in percentages. The values in parentheses are bootstrap (for *R*) and Newey-West (1987) adjusted (for α) *t*-statistics. The stars ***, **, and * indicate statistical significance the 1%, 5%, and 10% levels, respectively.

Table 3

Performance of the Value-Weighted Portfolios from One-Way Sorts on Past Alpha

	Countries							Industries						
	Bottom	2	3	4	Top	T-B	MR	Bottom	2	3	4	Top	T-B	MR
<i>Panel A: Alpha momentum</i>														
R	0.36 (1.20)	0.34 (1.39)	0.77*** (2.80)	0.87*** (3.70)	0.87*** (4.09)	0.51** (2.01)	2.58**	0.30 (1.15)	0.26 (1.09)	0.58*** (2.67)	0.80*** (3.64)	0.90*** (4.27)	0.61** (2.51)	8.03*
Vol	6.01	5.21	5.45	5.57	5.48	5.78		5.34	4.89	4.68	4.77	5.10	5.26	
SR	0.21	0.23	0.49	0.54	0.55	0.31		0.19	0.18	0.43	0.58	0.61	0.40	
α	-0.31* (-1.88)	-0.35*** (-2.75)	0.16 (1.16)	0.20* (1.66)	0.27* (1.83)	0.58** (2.25)	3.45**	-0.36*** (-2.58)	-0.38*** (-4.16)	0.01 (0.13)	0.27*** (3.22)	0.40*** (3.18)	0.76*** (3.27)	2.50**
MKT	1.02*** (21.83)	0.97*** (27.32)	1.03*** (28.53)	1.14*** (31.62)	0.98*** (19.11)	-0.04 (-0.54)		1.01*** (27.13)	1.04*** (39.08)	1.01*** (40.97)	1.00*** (30.76)	0.98*** (21.91)	-0.03 (-0.38)	
SMB	-0.01 (-0.09)	0.29*** (5.34)	0.08 (1.27)	0.22*** (3.74)	0.27*** (3.16)	0.28** (2.28)		0.02 (0.19)	0.01 (0.19)	0.10 (1.61)	0.15*** (2.88)	0.29*** (3.39)	0.27* (1.77)	
HML	0.28*** (4.82)	0.23*** (5.64)	0.07 (1.51)	-0.01 (-0.32)	-0.01 (-0.19)	-0.29*** (-4.00)		0.21*** (3.44)	0.14*** (2.81)	0.01 (0.16)	-0.10 (-1.33)	-0.19** (-2.49)	-0.40*** (-3.41)	
<i>Panel B: Alpha reversal</i>														
R	0.69** (2.44)	0.83*** (3.04)	0.69*** (2.62)	0.86*** (3.38)	0.51** (2.18)	-0.17 (-0.67)	38.81	0.67** (2.53)	0.68*** (2.91)	0.77*** (3.53)	0.66*** (2.99)	0.42* (1.76)	-0.25 (-1.18)	32.76
Vol	5.15	5.52	5.38	5.46	5.31	5.06		4.92	4.62	4.60	4.64	5.33	4.93	
SR	0.46	0.52	0.44	0.55	0.33	-0.12		0.47	0.51	0.58	0.49	0.27	-0.18	
α	0.10 (0.61)	0.07 (0.51)	-0.04 (-0.35)	0.18 (1.30)	-0.11 (-0.73)	-0.22 (-0.84)	43.75	0.05 (0.37)	0.04 (0.42)	0.12 (1.54)	0.05 (0.53)	-0.14 (-1.08)	-0.19 (-0.80)	33.49
MKT	0.91*** (22.12)	1.08*** (27.19)	1.04*** (32.02)	1.08*** (23.30)	1.02*** (19.13)	0.11 (1.38)		0.94*** (23.43)	0.98*** (38.76)	1.01*** (49.43)	1.01*** (37.17)	1.10*** (22.78)	0.16* (1.92)	
SMB	0.01 (0.13)	0.30*** (4.72)	0.17*** (3.74)	0.15*** (2.91)	0.07 (1.09)	0.06 (0.60)		0.11 (1.31)	0.09 (1.53)	0.10** (2.43)	0.04 (0.92)	-0.05 (-0.61)	-0.16 (-1.05)	
HML	0.15*** (2.81)	0.15*** (3.50)	0.25*** (5.78)	0.00 (-0.06)	0.02 (0.46)	-0.13* (-1.68)		0.08 (1.16)	0.09 (1.51)	0.06 (1.60)	0.03 (0.63)	-0.14* (-1.78)	-0.22 (-1.59)	

Note. The table reports the performance of value-weighted quintile portfolios of country (left-hand side) and industry (right-hand side) equity indexes from sorts on: the *alpha momentum*, i.e., the volatility-adjusted CAPM alpha estimated through months $t-12$ to $t-1$ (Panel A) and the *alpha reversal*, i.e., the volatility-adjusted CAPM alpha estimated through months $t-60$ to $t-13$ (Panel B). The *T-B* is the long-short portfolio going long in the *Top* and short in the *Bottom* portfolios. *R* is the mean monthly excess return; *Vol* is the standard deviation of monthly returns; and *SR* is the annualized Sharpe ratio. α is the alpha from the Fama-French (1993) three-factor model, and *MKT*, *SMB*, and *HML* are the measures of exposure of the market portfolio, small minus big, and high minus low factors, respectively. *MR* is the p -value from the Patton and Timmerman test of monotonic relationship. *R*, *Vol*, α , and *MR* are presented in percentages. The values in parentheses are bootstrap (for *R*) and Newey-West (1987) adjusted (for α) t -statistics. The stars ***, **, and * indicate statistical significance the 1%, 5%, and 10% levels, respectively.

Table 4

Performance of Long-Short Portfolios on Past Alphas Estimated from Different Factor Models

	<i>Panel A: Alpha momentum</i>				<i>Panel B: Alpha reversal</i>			
	Sorts on raw alphas		Sorts on volatility-adjusted alphas		Sorts on raw alphas		Sorts on volatility-adjusted alphas	
	Countries	Industries	Countries	Industries	Countries	Industries	Countries	Industries
<i>CAPM (1964)</i>								
R	0.87*** (3.33)	1.25*** (5.80)	0.84*** (3.75)	1.15*** (5.97)	-0.67*** (-2.97)	-0.59*** (-3.32)	-0.59*** (-2.81)	-0.63*** (-3.96)
Vol	5.76	4.73	5.03	4.28	4.50	3.54	4.29	3.29
SR	0.52	0.92	0.58	0.93	-0.52	-0.58	-0.48	-0.66
α	0.97*** (3.65)	1.54*** (5.95)	0.92*** (3.75)	1.41*** (5.98)	-0.63*** (-2.84)	-0.49*** (-2.76)	-0.52** (-2.33)	-0.49*** (-2.88)
<i>Asness, Frazzini, and Pedersen (2013) model</i>								
R	0.57*** (2.73)	0.90*** (4.70)	0.49*** (2.62)	0.91*** (5.16)	-0.68*** (-3.16)	-0.53*** (-3.49)	-0.60*** (-2.87)	-0.51*** (-3.61)
Vol	4.92	4.19	4.47	3.82	4.52	3.36	4.14	3.08
SR	0.40	0.74	0.38	0.83	-0.52	-0.55	-0.50	-0.57
α	0.70*** (3.24)	1.22*** (6.28)	0.57*** (2.73)	1.23*** (6.42)	-0.59*** (-2.85)	-0.33** (-1.98)	-0.54*** (-2.74)	-0.30* (-1.82)
<i>Fama and French (1993) model</i>								
R	0.70*** (3.54)	0.96*** (5.17)	0.66*** (2.99)	0.93*** (5.19)	-0.57** (-2.40)	-0.60*** (-3.53)	-0.68*** (-3.17)	-0.67*** (-4.14)
Vol	4.95	4.31	4.77	3.98	4.58	3.63	4.29	3.41
SR	0.49	0.77	0.48	0.81	-0.43	-0.57	-0.55	-0.68
α	0.82*** (4.68)	1.29*** (5.75)	0.78*** (3.68)	1.24*** (5.70)	-0.45** (-2.11)	-0.36** (-2.07)	-0.59*** (-2.77)	-0.42** (-2.53)
<i>Carhart (1997) model</i>								
R	0.49*** (2.58)	0.79*** (4.59)	0.42** (2.37)	0.87*** (5.38)	-0.60** (-2.44)	-0.50*** (-3.25)	-0.68*** (-3.12)	-0.55*** (-3.95)
Vol	4.77	3.99	4.45	3.73	4.58	3.33	4.20	3.04
SR	0.36	0.69	0.33	0.81	-0.45	-0.52	-0.56	-0.63
α	0.64*** (3.07)	1.09*** (6.10)	0.55*** (2.82)	1.19*** (6.37)	-0.45** (-2.37)	-0.28* (-1.79)	-0.59*** (-2.85)	-0.32** (-2.07)

Note. The table reports the performance of long-short equal-weighted quintile portfolios of country and industry equity indexes from sorts on: the *alpha momentum*, i.e., the alpha estimated through months $t-12$ to $t-1$ and the *alpha reversal*, i.e., the alpha estimated through months $t-60$ to $t-13$. The alphas are estimated using three different models: the three-factor model of Fama and French (1993), including the *MKT*, *SMB*, and *HML* factors; the four-factor model of Carhart (1997), including the *MKT*, *SMB*, *HML*, and *MOM* factors; and the three-factor model of Asness, Frazzini, and Pedersen (2013), including the *MKT*, *HML*, and *MOM* factors. The long-short portfolios go long in the *Top* and short in the *Bottom* alpha momentum or alpha reversal portfolio. *R* is the mean monthly excess return of the long-short portfolio; *Vol* is the standard deviation of monthly returns; *SR* is the annualized Sharpe ratio; and α is the alpha from the three-factor model of Fama and French (1993). *R*, *Vol*, and α are presented in percentages. The values in parentheses are bootstrap (for *R*) and Newey-West (1987) adjusted (for α) *t*-statistics. The stars ***, **, and * indicate statistical significance the 1%, 5%, and 10% levels, respectively.

Table 5

Influence of Trading Costs on Performance of Portfolios from One-Way Sorts on Past Alpha

	<u>Countries</u>				<u>Industries</u>			
	c=0.5%	c=1.0%	c=1.5%	c=2.0%	c=0.5%	c=1.0%	c=1.5%	c=2.0%
<i>Panel A: Alpha momentum effect: sorts in performance in months t-12 to t-1</i>								
R	0.64*** (2.89)	0.44** (2.03)	0.24 (1.17)	0.05 (0.31)	0.96*** (4.99)	0.76*** (4.01)	0.57*** (3.03)	0.38** (2.05)
Vol	5.04	5.05	5.06	5.08	4.29	4.29	4.30	4.31
SR	0.44	0.30	0.16	0.03	0.78	0.61	0.46	0.31
<i>Panel B: Alpha reversal effect: sorts in performance in months t-60 to t-13</i>								
R	0.49** (2.29)	0.38* (1.77)	0.27 (1.25)	0.17 (0.74)	0.53*** (3.32)	0.43*** (2.67)	0.33** (2.03)	0.23 (1.38)
Vol	4.28	4.27	4.27	4.26	3.29	3.29	3.28	3.28
SR	0.40	0.31	0.22	0.14	0.56	0.45	0.35	0.24

Note. The table reports the performance of equal-weighted long-short quintile portfolios of country (Panel A) and industry (Panel B) equity indexes from sorts on: the *alpha momentum* signal, i.e., the volatility-adjusted CAPM alpha estimated through months $t-12$ to $t-1$ and the *alpha reversal* signal, i.e., the volatility-adjusted CAPM alpha estimated through months $t-60$ to $t-13$. The long-short portfolios go long in the indexes with the highest and short in the ones with lowest alpha momentum or alpha reversal. *R* is the mean monthly excess return of the long-short portfolio, *Vol* is the standard deviation of monthly returns, and *SR* is the annualized Sharpe ratio. *R* and *Vol* are presented in percentages. The returns are adjusted with three different levels of one-way trading costs (*c*), amounting to 0.5%, 1%, 1.5%, and 2%. The values in parentheses are bootstrap *t*-statistics. The stars ***, **, and * indicate statistical significance the 1%, 5%, and 10% levels, respectively.

Table 6

Performance of Portfolios from One-Way Sorts on Past Alpha with Alternative Holding Periods

	<u>Countries</u>				<u>Industries</u>			
	R	t-stat _R	α	t-stat _{α}	R	t-stat _R	α	t-stat _{α}
<i>Panel A: Alpha momentum effect: sorts on performance in months t-12 to t-1</i>								
1 month	0.84***	(3.75)	0.92***	(3.75)	1.15***	(5.97)	1.41***	(5.97)
2 months	0.71***	(3.39)	0.82***	(3.52)	1.03***	(5.60)	1.30***	(5.67)
3 months	0.61***	(3.01)	0.70***	(3.08)	0.95***	(5.48)	1.21***	(5.33)
4 months	0.52***	(2.60)	0.62***	(2.81)	0.85***	(5.11)	1.11***	(5.08)
5 months	0.47**	(2.40)	0.56***	(2.60)	0.79***	(4.93)	1.04***	(4.82)
6 months	0.46**	(2.41)	0.56***	(2.66)	0.73***	(4.72)	0.98***	(4.78)
7 months	0.44**	(2.36)	0.53***	(2.62)	0.67***	(4.46)	0.92***	(4.57)
8 months	0.36**	(2.07)	0.45**	(2.36)	0.60***	(4.10)	0.84***	(4.41)
9 months	0.30*	(1.82)	0.39**	(2.04)	0.53***	(3.76)	0.77***	(4.17)
10 months	0.23	(1.47)	0.32*	(1.77)	0.45***	(3.30)	0.69***	(3.82)
11 months	0.17	(1.15)	0.27	(1.50)	0.39***	(2.92)	0.61***	(3.47)
12 months	0.11	(0.81)	0.21	(1.17)	0.32**	(2.50)	0.54***	(3.14)
<i>Panel B: Alpha reversal effect: sorts on performance in months t-60 to t-13</i>								
1 month	-0.59***	(-2.81)	-0.52**	(-2.33)	-0.63***	(-3.96)	-0.49***	(-2.88)
2 months	-0.62***	(-2.92)	-0.55**	(-2.53)	-0.65***	(-4.04)	-0.51***	(-3.05)
3 months	-0.64***	(-3.08)	-0.58***	(-2.70)	-0.68***	(-4.19)	-0.54***	(-3.28)
4 months	-0.63***	(-3.07)	-0.57***	(-2.67)	-0.68***	(-4.18)	-0.54***	(-3.36)
5 months	-0.61***	(-2.97)	-0.55***	(-2.65)	-0.67***	(-4.17)	-0.55***	(-3.42)
6 months	-0.59***	(-2.88)	-0.53***	(-2.59)	-0.66***	(-4.07)	-0.54***	(-3.39)
7 months	-0.59***	(-2.83)	-0.52***	(-2.62)	-0.65***	(-4.09)	-0.53***	(-3.41)
8 months	-0.58***	(-2.83)	-0.52***	(-2.64)	-0.65***	(-4.10)	-0.53***	(-3.46)
9 months	-0.57***	(-2.79)	-0.51***	(-2.63)	-0.64***	(-4.05)	-0.53***	(-3.49)
10 months	-0.54***	(-2.71)	-0.49**	(-2.56)	-0.64***	(-4.06)	-0.53***	(-3.54)
11 months	-0.55***	(-2.77)	-0.50***	(-2.64)	-0.65***	(-4.14)	-0.54***	(-3.65)
12 months	-0.55***	(-2.76)	-0.50***	(-2.64)	-0.65***	(-4.21)	-0.55***	(-3.74)

Note. The table reports the performance of equal-weighted long-short quintile portfolios of country (Panel A) and industry (Panel B) equity indexes from sorts on: the *alpha momentum* signal, i.e., the volatility-adjusted CAPM alpha estimated through months $t-12$ to $t-1$ and the *alpha reversal* signal, i.e., the volatility-adjusted CAPM alpha estimated through months $t-60$ to $t-13$. The long-short portfolios go long in the indexes with the highest and short in the ones with lowest alpha momentum or alpha reversal. The portfolios are rebalanced monthly and the first column indicates the length of the holding period. R is the mean monthly excess return of the long-short portfolio, α is the alpha from the three-factor model, and $t\text{-stat}_R$ and $t\text{-stat}_\alpha$ are corresponding bootstrap and Newey-West (1987) adjusted t -statistics. The stars ***, **, and * indicate statistical significance the 1%, 5%, and 10% levels, respectively.

Table 7

Past Alpha vs. Past Returns: Examination with the Fama-MacBeth Regressions

	Panel A: Countries				Panel B: Industries			
	Raw excess returns		Risk-adjusted returns		Raw excess returns		Risk-adjusted returns	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
AMOM	1.61** (2.23)		1.34** (1.98)		1.08*** (3.14)		1.03*** (3.06)	
AREV		0.14 (0.12)		0.07 (0.07)		-1.04** (-2.00)		-1.40*** (-2.78)
BET	0.00 (0.02)	-0.06 (-0.18)	-0.07 (-0.24)	-0.25 (-0.77)	0.01 (0.08)	0.00 (0.02)	-0.05 (-0.41)	-0.09 (-0.64)
MV	0.00 (-0.10)	0.04 (0.92)	0.00 (0.00)	0.05 (1.00)	-0.12*** (-5.26)	-0.06*** (-2.58)	-0.10*** (-4.86)	-0.05** (-2.17)
EP	8.92*** (3.21)	6.63*** (3.06)	7.01*** (2.67)	5.30*** (2.63)	2.91*** (3.92)	2.22*** (2.97)	2.66*** (4.11)	2.07*** (3.17)
IVOL	1.87 (0.36)	13.19** (2.19)	-0.50 (-0.10)	10.86 (1.61)	1.93 (0.95)	6.45*** (3.03)	-1.76 (-0.96)	2.69 (1.37)
SKEW	-0.02 (-0.16)	-0.05 (-0.36)	-0.03 (-0.22)	-0.02 (-0.14)	0.00 (0.02)	-0.03 (-0.43)	0.03 (0.44)	-0.01 (-0.14)
SEAS	1.27 (0.31)	2.88 (0.67)	-1.49 (-0.38)	-0.14 (-0.03)	5.12 (0.95)	5.32*** (3.39)	0.42 (0.28)	2.60 (1.63)
MOM	-0.54 (-0.51)		-0.47 (-0.49)		0.28** (2.30)		0.19 (0.43)	
REV		-0.50 (-1.37)		-0.48 (-1.40)		-0.10 (-0.85)		-0.15 (-1.31)
R ²	46.03	44.81	42.13	40.77	15.30	13.76	12.27	10.93

Note. This table reports the slope coefficients and the corresponding t-statistics from the monthly Fama-MacBeth cross-sectional regressions in the following form:

$$R_{i,t} = \beta_{0,t} + \sum_{j=1}^J \beta_{j,t} K_{i,t} + \varepsilon_{i,t},$$

where $R_{i,t}$ is the excess return (Panels A.1 and B.1) or three-factor model-adjusted returns (Panels A.2 and B.2) on index i in month t ; $\beta_{0,t}$ and $\beta_{j,t}$ are regression parameters; and $K_{i,t}$ are various return predictors: volatility-adjusted alpha from the CAPM estimated through the months $t-12$ to $t-1$ (AMOM); volatility-adjusted alpha from the CAPM estimated through the months $t-60$ to $t-13$ (AREV); stock market beta (BETA) and idiosyncratic volatility (IVOL) estimated with the CAPM through the months $t-60$ to $t-1$; log-market value at month $t-1$ (MV); and earnings-to-price ratio at month $t-1$ (EP); skewness of the return distribution estimated through the months $t-24$ to $t-1$ (SKEW); and average return in the same calendar month as t through the months $t-240$ to $t-1$, as available (SEAS); momentum (MOM), i.e., the total return in months $t-12$ to $t-1$; and reversal (REV), i.e., the total return in months $t-60$ to $t-13$. R^2 is the average cross-sectional coefficient of determination (expressed in %). The values in parentheses are Newey-West (1987) adjusted t -statistics. The stars ***, **, and * indicate statistical significance the 1%, 5%, and 10% levels, respectively. The table's sections display the results based on country (Panel A) and industry (Panel B) indexes.

Table 8

Results of the Factor Spanning Tests

	α	MKT	SMB	HML	MOM	REV	AMOM	AREV	R ²
<i>Panel A: Countries</i>									
AMOM	0.95*** (4.39)	0.06 (0.96)	0.20* (1.94)	-0.34*** (-4.94)					9.53
AMOM	0.42*** (3.19)	0.08 (1.36)	0.08 (1.47)	-0.03 (-0.58)	0.68*** (12.67)				59.13
AMOM	0.43*** (3.20)	0.03 (0.60)	0.09* (1.69)	-0.09 (-1.64)	0.70*** (19.46)	-0.01 (-0.31)			62.03
MOM	0.01 (0.05)	-0.07 (-1.10)	0.01 (0.21)	-0.17* (-1.84)			0.81*** (21.22)		60.63
MOM	-0.05 (-0.32)	-0.01 (-0.23)	0.03 (0.47)	-0.08 (-1.47)			0.82*** (19.28)	0.02 (0.44)	61.32
AREV	0.43** (2.08)	-0.06 (-1.21)	-0.11 (-1.50)	0.16** (2.46)					2.76
AREV	0.27* (1.85)	-0.04 (-0.97)	-0.01 (-0.12)	0.04 (0.73)		0.62*** (12.13)			41.19
AREV	0.25* (1.77)	-0.04 (-0.99)	-0.01 (-0.24)	0.05 (0.90)	0.03 (0.71)	0.62*** (12.32)			41.18
REV	-0.01 (-0.09)	0.00 (0.10)	-0.10* (-1.73)	0.09* (1.84)				0.64*** (11.00)	42.23
REV	0.03 (0.20)	0.01 (0.15)	-0.09 (-1.54)	0.08 (1.42)			-0.04 (-0.97)	0.64*** (11.14)	42.33
<i>Panel B: Industries</i>									
AMOM	0.99*** (5.24)	-0.04 (-0.76)	0.17 (1.46)	-0.38*** (-4.67)					7.50
AMOM	0.21** (2.02)	0.09** (2.15)	0.15*** (2.89)	-0.02 (-0.37)	0.71*** (17.42)				73.87
AMOM	0.28*** (2.66)	0.07 (1.62)	0.16*** (2.77)	-0.03 (-0.68)	0.70*** (13.79)	-0.04 (-0.83)			72.04
MOM	0.10 (1.07)	-0.14*** (-2.62)	-0.15*** (-3.03)	-0.12 (-1.64)			1.01*** (31.51)		75.09
MOM	0.06 (0.56)	-0.11* (-1.91)	-0.12** (-2.34)	-0.12 (-1.49)			0.99*** (30.20)	0.05 (0.83)	72.27
AREV	0.21* (1.71)	-0.03 (-0.66)	0.02 (0.28)	0.23*** (2.65)					3.98
AREV	0.09 (0.92)	-0.01 (-0.35)	-0.01 (-0.15)	0.08 (1.34)		0.70*** (17.08)			62.51
AREV	0.01 (0.05)	0.00 (-0.07)	-0.02 (-0.38)	0.11* (1.78)	0.07** (2.23)	0.71*** (19.46)			63.39
REV	-0.01 (-0.08)	0.00 (0.04)	0.02 (0.36)	0.02 (0.27)				0.87*** (17.68)	62.07
REV	0.10 (0.92)	0.00 (-0.01)	0.05 (0.74)	-0.02 (-0.38)			-0.10*** (-2.65)	0.86*** (17.76)	63.10

Note. The table displays the results of the time-series spanning tests of the alpha momentum and alpha reversal factor portfolios. All the factor portfolios are constructed using independent sorts on country (Panel A) and industry (Panel B) indexes to create two size-related and three variable-related groups. The table shows the slope coefficients from the regressions of the alpha momentum and reversal factors on the sets of other factors. The alpha momentum signal (AMOM) is the volatility-adjusted CAPM alpha estimated through months $t-12$ to $t-1$, and the alpha reversal (AREV) signal is the volatility-adjusted CAPM alpha estimated through months $t-60$ to $t-13$. The remaining factors considered include market excess return (MKT), small minus big (SMB), high minus low (HML), momentum (MOM), and reversal (REV). α is the regression intercept and R^2 is the coefficient of determination, both presented in percentages. The values in parentheses are Newey-West (1987) adjusted t -statistics. The stars ***, **, and * indicate statistical significance the 1%, 5%, and 10% levels, respectively.

Table 9

Abnormal Returns on Portfolios from Two-Way Dependent Sorts on Past Performance

Countries					Industries				
Panel A: Sorts on alphas within return quantiles									
AMOM									
	Low	Medium	High	High-Low	Low	Medium	High	High-Low	
MOM	High	0.22 (1.35)	0.34** (1.98)	0.55*** (2.71)	0.33* (1.78)	0.47*** (4.28)	0.69*** (5.51)	1.21*** (6.68)	0.73*** (4.13)
	Medium	-0.21 (-1.45)	0.01 (0.07)	0.31** (2.31)	0.52*** (2.88)	-0.11 (-1.18)	-0.06 (-0.84)	0.30*** (2.72)	0.41*** (2.85)
	Low	-0.42** (-1.98)	-0.29* (-1.76)	-0.34* (-1.95)	0.08 (0.28)	-0.57*** (-3.62)	-0.44*** (-3.65)	-0.16 (-1.40)	0.41*** (2.77)
AREV									
	Low	Medium	High	High-Low	Low	Medium	High	High-Low	
REV	High	0.26* (1.83)	-0.12 (-0.76)	-0.39** (-2.18)	-0.65*** (-2.77)	0.15 (1.48)	-0.04 (-0.39)	-0.29** (-2.48)	-0.44*** (-3.10)
	Medium	0.06 (0.49)	-0.06 (-0.42)	-0.01 (-0.07)	-0.07 (-0.39)	0.20** (2.08)	0.18** (2.11)	0.03 (0.38)	-0.17* (-1.69)
	Low	0.25 (1.23)	0.25 (1.45)	0.07 (0.39)	-0.17 (-0.62)	0.41*** (2.88)	0.09 (0.72)	0.29* (1.90)	-0.13 (-0.66)
Panel B: Sorts on returns within alpha quantiles									
MOM									
	Low	Medium	High	High-Low	Low	Medium	High	High-Low	
AMOM	High	0.57*** (3.26)	0.40*** (2.81)	0.48** (2.37)	-0.10 (-0.40)	0.42*** (3.71)	0.61*** (5.92)	1.27*** (6.84)	0.85*** (4.71)
	Medium	-0.10 (-0.58)	0.00 (0.00)	-0.08 (-0.55)	0.02 (0.09)	-0.08 (-0.75)	0.01 (0.15)	0.33*** (2.96)	0.41 (1.51)
	Low	-0.32 (-1.34)	-0.70*** (-5.25)	-0.08 (-0.55)	0.24 (0.93)	-0.46** (-2.36)	-0.59*** (-5.28)	-0.19* (-1.69)	0.27 (1.33)
REV									
	Low	Medium	High	High-Low	Low	Medium	High	High-Low	
AREV	High	0.03 (0.20)	-0.17 (-1.02)	-0.35* (-1.88)	-0.38 (-1.50)	0.07 (0.70)	-0.04 (-0.54)	-0.22* (-1.68)	-0.29 (-1.56)
	Medium	-0.08 (-0.44)	-0.02 (-0.14)	0.18 (1.23)	0.26 (1.13)	0.23* (1.81)	0.14 (1.43)	0.14 (1.36)	-0.08 (-0.52)
	Low	0.32 (1.34)	0.13 (0.75)	0.33** (2.31)	0.01 (0.05)	0.44** (2.55)	0.08 (0.80)	0.21** (2.26)	-0.23 (-1.18)

Note. The table displays the three-factor model alphas of the portfolios of countries (left-hand side) and industries (right-hand side) from two-way dependent sorts on measures of past performance: alpha momentum (AMOM), i.e., the volatility-adjusted CAPM alpha estimated through months $t-12$ to $t-1$; alpha reversal (AREV), i.e., the volatility-adjusted CAPM alpha estimated through months $t-60$ to $t-13$; momentum (MOM), i.e., the total return in months $t-12$ to $t-1$; and reversal (REV), i.e., the total return in months $t-60$ to $t-13$. The alphas are presented in percentages. The values in parentheses are Newey-West (1987) adjusted t -statistics. The stars ***, **, and * indicate statistical significance the 1%, 5%, and 10% levels, respectively.

Table 10

Abnormal Returns on Portfolios from Two-Way Sorts on Past Alpha and Additional Variables

		Countries				Industries			
		Panel A: Alpha momentum: performance in months $t-12$ to $t-1$							
		Low	Medium	High	High-Low	Low	Medium	High	High-Low
Panel A.1: Two-way sort on market value									
Sorts on market value	High	-0.29*** (-2.66)	0.08 (0.68)	-0.04 (-0.30)	0.25 (1.34)	-0.37*** (-3.82)	-0.02 (-0.34)	0.33*** (3.18)	0.70*** (3.96)
	Medium	-0.44** (-2.07)	-0.08 (-0.49)	0.43** (2.54)	0.87*** (3.01)	-0.61*** (-4.87)	-0.12 (-1.52)	0.48*** (4.07)	1.09*** (5.04)
	Low	-0.15 (-0.77)	-0.06 (-0.34)	0.72*** (3.25)	0.87*** (3.04)	-0.20 (-1.39)	0.41*** (4.01)	1.43*** (8.91)	1.63*** (6.83)
Panel A.2: Two-way sort on idiosyncratic risk									
Sorts on idiosyncratic risk	High	-0.20 (-0.74)	-0.03 (-0.16)	0.55*** (3.02)	0.76** (2.38)	-0.36* (-1.83)	0.23* (1.68)	0.88*** (4.48)	1.24*** (3.83)
	Medium	-0.19 (-1.33)	-0.08 (-0.45)	0.20 (1.21)	0.38* (1.71)	-0.43*** (-2.96)	0.00 (-0.06)	0.68*** (5.87)	1.11*** (5.13)
	Low	-0.13 (-1.12)	0.01 (0.13)	0.14 (1.28)	0.27* (1.81)	-0.29*** (-3.13)	-0.02 (-0.24)	0.34*** (3.50)	0.63*** (5.04)
Panel A.3: Two-way sort on R^2 coefficient									
Sorts on R^2 coefficient	High	-0.35** (-2.03)	-0.01 (-0.07)	0.30** (2.16)	0.65*** (2.92)	-0.39** (-2.50)	-0.10 (-1.23)	0.40*** (4.06)	0.79*** (3.73)
	Medium	-0.27 (-1.50)	-0.03 (-0.22)	0.24 (1.51)	0.51** (2.06)	-0.42*** (-3.20)	0.03 (0.36)	0.59*** (5.12)	1.01*** (4.57)
	Low	-0.16 (-0.78)	-0.10 (-0.51)	0.78*** (3.45)	0.94*** (2.90)	-0.24* (-1.82)	0.41*** (3.26)	1.29*** (6.91)	1.53*** (6.36)
Panel B: Alpha reversal: performance in months $t-60$ to $t-13$									
		Low	Medium	High	High-Low	Low	Medium	High	High-Low
Panel B.1: Two-way sort on market value									
Sorts on market value	High	-0.01 (-0.07)	-0.03 (-0.32)	-0.15 (-1.13)	-0.14 (-0.78)	0.01 (0.08)	0.06 (0.93)	-0.11 (-1.22)	-0.11 (-0.72)
	Medium	0.17 (0.99)	0.16 (0.88)	-0.29 (-1.55)	-0.46* (-1.68)	0.08 (0.73)	0.01 (0.12)	-0.25*** (-2.66)	-0.32** (-2.05)
	Low	0.54** (2.55)	0.15 (0.79)	-0.12 (-0.64)	-0.66** (-2.30)	0.55*** (4.07)	0.42*** (3.79)	0.26*** (2.73)	-0.29** (-1.96)
Panel B.2: Two-way sort on idiosyncratic risk									
Sorts on idiosyncratic risk	High	0.54** (2.33)	0.16 (0.69)	-0.34* (-1.75)	-0.87*** (-2.98)	0.48*** (2.96)	0.38** (2.50)	-0.10 (-0.70)	-0.58*** (-2.76)
	Medium	0.18 (1.21)	0.09 (0.69)	-0.23 (-1.55)	-0.41** (-2.05)	0.26*** (2.75)	0.03 (0.38)	-0.05 (-0.55)	-0.30** (-2.15)
	Low	0.04 (0.40)	0.02 (0.19)	-0.05 (-0.40)	-0.09 (-0.60)	-0.06 (-0.57)	0.07 (1.01)	0.01 (0.11)	0.07 (0.57)
Panel B.3: Two-way sort on R^2 coefficient									
Sorts on R^2 coefficient	High	-0.10 (-0.87)	-0.01 (-0.05)	-0.04 (-0.34)	0.06 (0.31)	-0.11 (-0.87)	0.02 (0.17)	-0.07 (-0.69)	0.04 (0.26)
	Medium	0.20 (1.02)	0.00 (-0.03)	-0.29* (-1.79)	-0.49** (-1.99)	0.20** (2.06)	0.07 (0.93)	-0.25** (-2.44)	-0.45*** (-2.86)
	Low	0.55*** (2.65)	0.19 (1.24)	-0.13 (-0.66)	-0.68** (-2.35)	0.66*** (5.18)	0.48*** (3.52)	0.02 (0.20)	-0.64*** (-3.84)

Note. The table reports the three-factor model alphas of equal-weighted portfolios of country (left side) and industry (right side) equity indexes from two-way dependent sorts on past performance and additional variables. The additional variables used for sorting in the first pass include the natural logarithm of market value at $t-1$, and the idiosyncratic volatility (*IVOL*) and the R^2 coefficient (R^2) from the CAPM estimated through months $t-60$ to $t-1$. The measures of past performance used to sort indexes in the second pass are the alpha momentum (*AMOM*), i.e., the volatility-adjusted CAPM alpha estimated through months $t-12$ to $t-1$; and the alpha reversal (*AREV*), i.e., the volatility-adjusted CAPM alpha estimated through months $t-60$ to $t-13$. The *High-Low* is the long-short portfolio going long in the *High* and short in the *Low* *AMOM* (*AREV*) portfolio. The alphas are presented in percentages. The values in parentheses are Newey-West (1987) adjusted t -statistics. The stars ***, **, and * indicate statistical significance the 1%, 5%, and 10% levels, respectively.

Table 11

Performance on Portfolios Formed on Past Alphas in Developed and Emerging Markets

		<u>Country Indexes</u>				<u>Industry Indexes</u>			
		Low	Medium	High	High-Low	Low	Medium	High	High-Low
<i>Panel A: Alpha momentum: sorts performance in months t-12 to t-1</i>									
Developed markets	R	0.39 (1.46)	0.71*** (3.02)	0.89*** (3.95)	0.50*** (2.97)	0.40 (1.57)	0.81*** (3.53)	1.26*** (5.99)	0.85*** (6.02)
	α	-0.33*** (-3.34)	0.05 (0.40)	0.24* (1.93)	0.57*** (3.50)	-0.46*** (-4.59)	0.05 (0.53)	0.59*** (5.17)	1.05*** (6.51)
Emerging markets	R	0.65 (1.57)	0.93** (2.43)	1.34*** (4.11)	0.69** (2.41)	1.07*** (3.43)	1.10*** (4.09)	1.94*** (5.95)	0.87*** (3.08)
	α	-0.25 (-0.82)	-0.01 (-0.05)	0.53*** (3.24)	0.79** (2.20)	-0.04 (-0.15)	0.08 (0.42)	1.02*** (4.57)	1.06*** (2.99)
<i>Panel B: Alpha reversal: sorts performance in months t-60 to t-13</i>									
Developed markets	R	0.84*** (3.12)	0.85*** (3.60)	0.55** (2.24)	-0.29* (-1.78)	1.00*** (3.87)	0.96*** (4.08)	0.69*** (3.11)	-0.31** (-2.42)
	α	0.12 (1.06)	0.10 (0.94)	-0.16 (-1.42)	-0.28* (-1.90)	0.14 (1.37)	0.16* (1.92)	-0.07 (-0.76)	-0.21* (-1.70)
Emerging markets	R	1.37*** (3.25)	0.79** (2.07)	0.56 (1.36)	-0.81** (-2.37)	1.37*** (3.80)	1.28*** (3.75)	0.85*** (2.72)	-0.53** (-2.40)
	α	0.36** (1.97)	-0.11 (-0.41)	-0.35 (-1.55)	-0.70** (-2.29)	0.30* (1.67)	0.19 (0.94)	-0.10 (-0.41)	-0.40* (-1.74)

Note. The table reports the performance of equal-weighted tertile portfolios of country (left-hand side) and industry (right-hand side) equity indexes from sorts on: the *alpha momentum* (Panel A), i.e., the volatility-adjusted CAPM alpha estimated through months $t-12$ to $t-1$ (Panel A) and the *alpha reversal* (Panel B), i.e., the volatility-adjusted CAPM alpha estimated through months $t-60$ to $t-13$ (Panel B). The strategies are implemented separately in developed and emerging markets. The *High-Low* portfolio is the long-short portfolio going long in the *High* and short in the *Low* portfolio. R is the mean monthly excess return and α is the three-factor model. R and α are presented in percentages. The values in parentheses are bootstrap (for R) and Newey-West (1987) adjusted (for α) t -statistics. The stars ***, **, and * indicate statistical significance the 1%, 5%, and 10% levels, respectively.

Table 12

Returns on the Portfolios Formed on Past Alphas in Subperiods

	<i>Panel A: Alpha momentum</i>				<i>Panel B: Alpha reversal</i>			
	Countries		Industries		Countries		Industries	
	R	t-stat	R	t-stat	R	t-stat	R	t-stat
First half	0.98***	(2.88)	1.07***	(3.77)	-0.92***	(-3.15)	-0.79***	(-3.53)
Second half	0.69**	(2.49)	1.23***	(4.92)	-0.33	(-1.17)	-0.50**	(-2.32)
High VIX	0.66*	(1.72)	1.05***	(3.23)	-0.80**	(-2.31)	-0.86***	(-3.28)
Low VIX	0.66**	(1.99)	1.19***	(4.24)	0.06	(0.28)	-0.09	(-0.26)
High Credit	0.52*	(1.65)	0.77***	(3.00)	-0.85***	(-2.86)	-0.89***	(-3.46)
Low Credit	1.14***	(4.42)	1.52***	(6.08)	-0.33	(-1.30)	-0.37**	(-1.98)
High TED	0.79**	(2.06)	1.01***	(3.10)	-0.57*	(-1.72)	-0.57**	(-2.23)
Low TED	0.53	(1.54)	1.23***	(4.47)	-0.17	(-0.46)	-0.39	(-1.34)
High BW	0.91***	(3.00)	1.17***	(4.34)	-0.70**	(-2.47)	-0.73***	(-3.36)
Low BW	0.92***	(2.84)	1.20***	(4.44)	-0.49*	(-1.71)	-0.54**	(-2.19)
High MICE	0.69**	(2.26)	1.29***	(4.88)	-0.79***	(-2.59)	-0.73***	(-3.25)
Low MICE	1.02***	(3.03)	1.10***	(3.84)	-0.40	(-1.48)	-0.56***	(-2.60)
High ECES	0.94***	(2.95)	1.36***	(4.86)	-0.90***	(-2.94)	-0.68***	(-3.20)
Low ECES	0.76**	(2.39)	0.94***	(3.79)	-0.31	(-1.13)	-0.60***	(-2.68)
January	0.49	(0.50)	1.42	(1.61)	-1.69*	(-1.69)	-1.83***	(-2.72)
Non-January	0.87***	(4.05)	1.12***	(6.32)	-0.50**	(-2.31)	-0.52***	(-3.14)
Bull mkt. (24 m.)	0.95***	(3.57)	1.36***	(5.88)	-0.60***	(-2.59)	-0.56***	(-3.13)
Bear mkt. (24 m.)	0.44	(1.11)	0.48	(1.59)	-0.59	(-1.37)	-0.89***	(-2.62)
Bull mkt. (36 m.)	1.01***	(3.97)	1.38***	(6.05)	-0.67***	(-2.86)	-0.64***	(-3.66)
Bear mkt. (36 m.)	0.48	(1.11)	0.74**	(1.97)	-0.31	(-0.86)	-0.59*	(-1.95)
Bull mkt. (48 m.)	0.95***	(3.77)	1.31***	(5.65)	-0.63***	(-2.70)	-0.65***	(-3.92)
Bear mkt. (48 m.)	0.52	(1.23)	0.86**	(2.52)	-0.45	(-1.16)	-0.54	(-1.45)

Note. The table reports the returns on long-short equal-weighted quintile portfolios of the country and industry equity indexes within subperiods. The portfolios go long in the indexes with the highest and short in the ones with the lowest *alpha momentum* (Panel A), i.e., the volatility-adjusted CAPM alpha estimated through months $t-12$ to $t-1$, or the *alpha reversal* (Panel B), i.e., the volatility-adjusted CAPM alpha estimated through months $t-60$ to $t-13$. R is the mean monthly excess return expressed in %. The values in brackets are Newey-West (1987) adjusted t -statistics. The stars ***, **, and * indicate statistical significance the 1%, 5%, and 10% levels, respectively. *First half* and *second half* refer to the periods January 1972–March 1995 and April 1995–May 2018, respectively. *VIX* is the VIX volatility index, *Credit* is the BAA credit spread, *TED* is the TED spread, *BW* is the Baker and Wurgler (2006) sentiment index, *MICE* is the University of Michigan Index of Consumer Expectations, and *ECES* is the European Commission's Economic Sentiment Indicator. *January* and *Non-January* show returns in January and in the other calendar months, respectively. *Bull mkt.* (*Bear mkt.*) indicates the performance in months following periods of positive (negative) cumulative return in the past 24, 36, or 48 months.